



MATHS in The Wild



Denis Day



Quality ready-to-use
resources



MATHS in The Wild



Denis Day

This book was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover and David Dickson supplied the illustrations.

Denis Day completed a mechanical engineering diploma and worked as a production engineer before completing his education qualifications (Master of Education). For more than 30 years he has been teaching maths and science in socially disadvantaged schools and he has also taught in private schools. Denis has presented numerous workshops at the Maths Association of Victoria's annual maths conference and regional conferences, and developed many curriculum materials.

His particular strength is in devising high interest activities which immediately provide a meaningful context for the mathematical concepts being studied – making learning both more effective and much more enjoyable for students.

Denis's interests include playing and coaching tennis (Level 2 tennis coach), saltwater fishing and Aussie Rules football (Level 2 umpire).

© Denis Day and Blake Education 2009

Downloadables sold with a licence to print and photocopy

The contents of this book are covered by copyright laws in Australia and in other countries.

The purchase price includes the right for you to print and photocopy these pages for your students without further payment to the publisher. For further information about how schools can use this photocopiable resource, please refer to the 'FAQ's on www.blake.com.au.

Otherwise, you may only download, print, copy, store and communicate the contents of this book without permission when that is allowed by the copyright laws applicable to you.

If you work in an Australian school that is covered by the education statutory licence, you may be allowed to copy, download, print, store and communicate parts of this book without permission from copyright owners. Under the education statutory licence, payments are made on behalf of schools to Copyright Agency, which distributes the payments to writers, artists and publishers. See www.copyright.com.au.

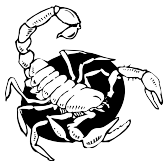
Copyright Agency
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

Printed in Australia by Ligare Book Printer

Blake
EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

ISBN 978 1 92571 009 0



Maths in the Wild contains photocopiable worksheets about some of the most interesting creatures that are found in our world’s saltwater, freshwater and land environments, such as whales, seahorses, snakes, tigers, mosquitoes, bees, eagles, cane toads and swans. The worksheets also examine pests introduced into the wild and the impact of humans on a wide range of species.

Each unit of work has been written to challenge and engage students in years 9 to 10, and contains a variety of content such as basic operations, measurement, graphs, percentages and analysis of data. Throughout these activities, students will also practise and extend their literacy skills as they identify the relevant data in information and encounter challenging contextual terms such as ‘ovoviviparous’, which are all clearly explained. Most pages include several list words for which students are asked to find the meaning.

Contents

1 Bottled, spotted and spinners – dolphins 4

2 Fish ‘n’ chips – sharks 6

3 Giants of the ocean – whales 8

4 Mighty molluscs – octopus, squid and cuttlefish..... 10

5 Flowers of the sea – anemones and jellyfish 12

6 Marinara on the menu – oysters and scallops..... 14

7 Weedy dragons and wet horses – seahorses 16

8 Exciting echinoderms – starfish 18

9 Native fish – Golden Perch and Murray Cod 20

10 A cute crustacean – yabbies 22

11 The Camelidae family – camels 24

12 Silent and deadly – snakes and scorpions 26

13 Endangered big cat – tigers 28

14 Queens and kings of the jungle – lions 30

15 Stings and wings – bees and wasps 32

16 Blood suckers – mosquitoes 34

17 Eagle eyes – birds of prey 36

18 Pesky visitors – rabbits and foxes..... 38

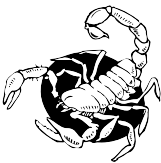
19 Big Bufo Marinus – Cane Toads 40

20 Graceful gliders – pelicans and swans 42

21 In the wild puzzles 44

Answers..... 46

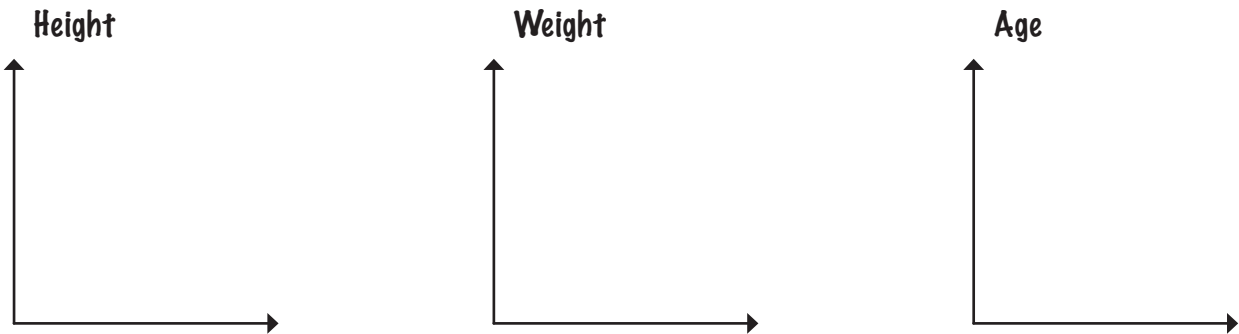
Bottled, spotted and spinners - dolphins



Dolphins live in the oceans and are closely related to whales and porpoises. There are almost 40 species of dolphin. They are found worldwide, mostly in the shallower seas of the continental shelves. Dolphins are carnivores, and eat mostly fish and squid. They are considered to be among the most intelligent of animals, and their often friendly appearance and seemingly playful attitude have made them popular with humans.

- 1 Find the meaning of each word listed below.
- a** carnivore **b** breach **c** sloughed **d** conical

- 2 Bottlenose Dolphins are on average 3.6 m long, weigh 500 kg and live for 45 years. An average human is 1.72 m high, 75 kg and lives for 80 years. Draw three column graphs to compare the average dolphin with the average human. Think carefully about the horizontal and vertical scales in your graphs.



- 3 Draw a picture graph to show the data in the table, which compares the different species of dolphins.

Type of dolphin	Length (m)	Weight (kg)
Bottlenose	2.7	320
Spotted	1.9	110
Spinner	1.6	72
Amazonian River	1.5	45
Orca (Killer Whale)	9.5	800



- 4 Bottlenose Dolphins can breach 5 metres out of the water, landing with a splash on their back or side. Change these breach heights to centimetres.
- a** 2 m **b** 3 m **c** 2.6 m
- d** 3.9 m **e** 2.69 m **f** 0.67 m
- 5 Dolphins normally cruise at speeds of about 8 km/h. Calculate how far a dolphin could swim at a speed of 8 km/h in each time listed below.
- a** 2 hours **b** 5 hours **c** 6 hours
- d** 3.4 hours **e** 4 h 30 min **f** 2 h 20 min
- 6 In the wild, these sleek swimmers can reach speeds of over 30 km/h (approximately 8 m/s) while chasing food. Calculate the distance a dolphin travelling at 9 m/s could swim in each time listed below. Answer in metres.
- a** 3 s **b** 5 s **c** 7 s
- d** 2 min **e** 4 min **f** 2 min 35 s

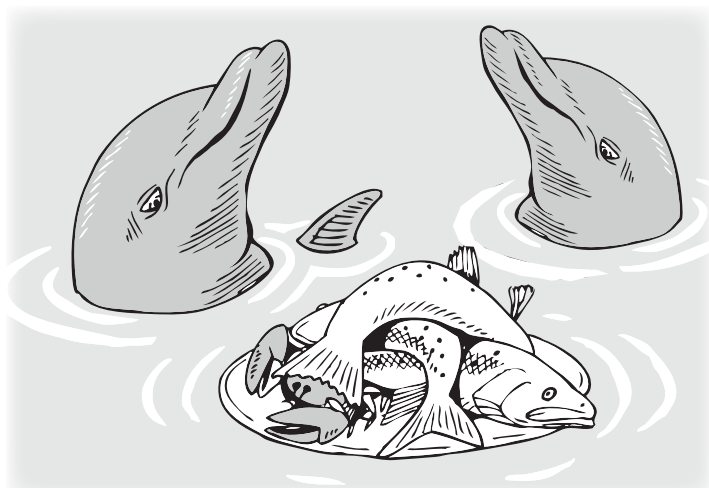


- 7** Dolphins surface often to breathe, approximately three times every minute. At a breathing rate of 3 times per minute, how many times would a dolphin breathe during each time listed below?

a 10 min **b** 5 min **c** 7 min
d 2 hours **e** 3 hr 21 min **f** 180 sec

- 8** An adult Bottlenose Dolphin eats on average 12 kg of food each day. They eat a variety of food, including fish, squid and crabs. Calculate the average quantity of food these numbers of dolphins would eat in one day.

a 10 dolphins
b 15 dolphins
c 23 dolphins
d 54 dolphins
e 63 dolphins
f 213 dolphins



- 9** A large salmon weighs approximately 2 kg. How many 2 kg salmon would these groups of dolphins need to eat in one day, to meet their average daily need of 12 kg per dolphin?

a 12 dolphins **b** 8 dolphins **c** 21 dolphins
d 25 dolphins **e** 55 dolphins **f** 120 dolphins

- 10** Convert each percentage to a decimal and then to a fraction in simplest form.

a 23% **b** 30% **c** 45% **d** 4% **e** 7%

- 11** On average, dolphins live for 20 years. Almost 1.5% reach the maximum age of 46 years. In each of these pods, how many dolphins would you expect to live to the maximum age?

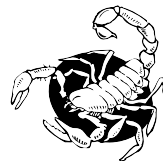
a 150 dolphins **b** 70 dolphins **c** 250 dolphins
d 600 dolphins **e** 470 dolphins **f** 680 dolphins

- 12** Suggest why so few dolphins reach the maximum age.

- 13** The smooth skin of a dolphin reduces its drag in the water and thus increases its swimming speed. The skin is kept smooth by constantly being sloughed off and replaced. A Bottlenose Dolphin replaces its outermost layer of skin every two hours. This is nine times the rate of human skin renewal. Calculate the number of times a dolphin sheds its outermost layer of skin during each time listed.

a 8 hours **b** 12 hours **c** 18 hours
d 4 days **e** 2 weeks **f** 1 year

- 14** How many times during one week would a human shed their outermost layer of skin?



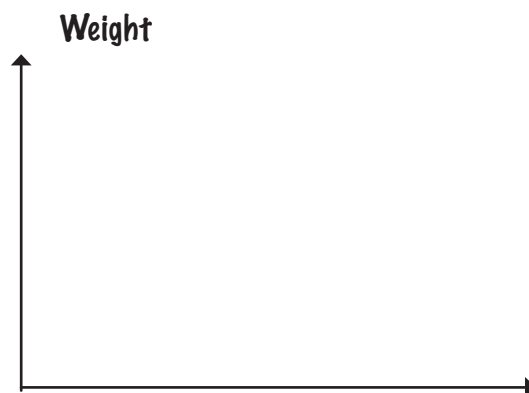
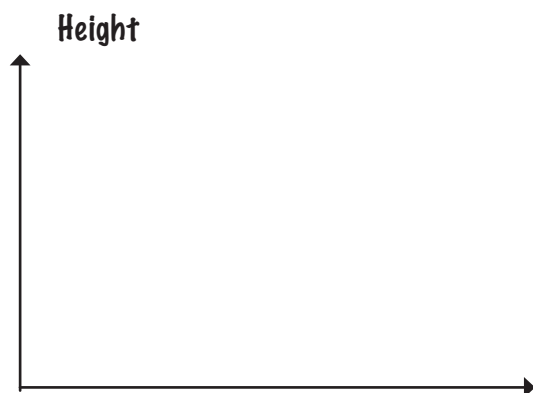
Sharks are a type of fish with a full cartilaginous skeleton and a streamlined body. They breathe with the use of five to seven gill slits. Sharks have a covering of dermal denticles (the skin feels like sandpaper) to protect their skin from damage and from parasites and to improve fluid dynamics. They also have replaceable teeth.

1 Find the meaning of each word listed below.

a cartilaginous **b** dermal **c** denticle **d** embryos **e** predatory

Sharks range in size from the small Spined Pygmy Shark, a deep sea species of only 22 centimetres in length, to the Whale Shark, the largest fish, which grows to a length of approximately 12 metres and 25 tonnes. The White Pointer grows to 7 m and 2 tonnes, Tiger Sharks to 5 m and 900 kg, Thresher Sharks to 5 m and 0.4 tonnes and Basking Sharks to 12.3 m and 19 tonnes. Humans can be up to 272 cm tall and 310 kg in weight.

2 Draw column graphs to compare the height and weight of a human with those of the given sharks.



The teeth of carnivorous sharks are not attached to the jaw, but embedded in the flesh. In many species, the teeth are constantly replaced throughout the shark's life. Sharks have about 3 000 teeth spread over 5 rows. The front set is the largest and does most of the work. In some sharks, a row of teeth is replaced every two weeks. Sharks live for approximately 25 years.

3 **a** Approximately how many teeth does a shark have in a single row?

b How many teeth are replaced every four weeks?

4 Calculate the approximate number of teeth each shark may have in its lifetime, if it lives for:

a 5 years. **b** 12 years.

c 2 years. **d** 15 years.

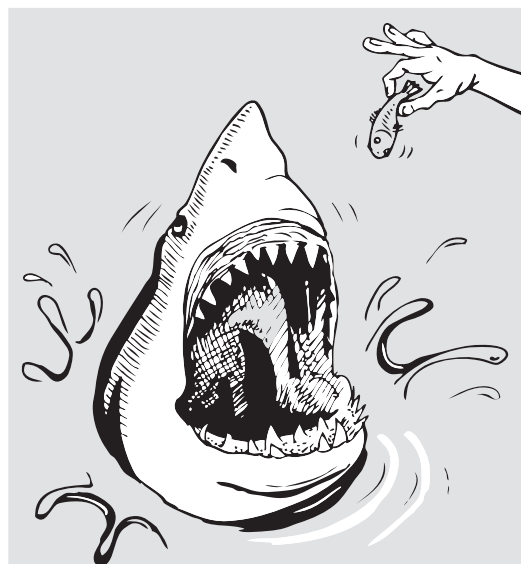
e 25 years.

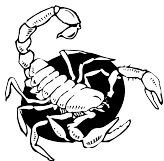
Sharks rarely attack humans unless provoked. In 2006, there were approximately 98 shark attacks on humans. Of these attacks, 64 were unprovoked and 18 were provoked, with the remainder uncertain.

5 Calculate the approximate percentage of attacks that were:

a provoked.

b unprovoked.

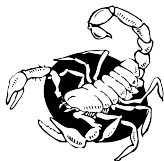




The table below shows the number of fatalities in Australia over a 10 year period from different activities.

Activity	Deaths over 10 yrs	Average deaths per year	Percentage deaths per year	Parts / 360°
Crocodile attack	9			
Shark attack	13			
Lightning strike	21			
Bee sting	25	2.5	0.07	0.25
Scuba diving	89			
Drowning	3 400			
Motor vehicle accident	32 900			
Total	36 457			

- 6 Complete the table by calculating:
- a the average yearly death rate.
 - b the percentage of deaths per year.
 - c parts per 360°.
- 7
- a Draw a large sector graph on a separate sheet of paper to display the data.
 - b Is this graph suitable to display the data? Explain your answer.
- 8 Great Whites account for about five-twelfths of all annual reported shark attacks. Of these Great White attacks, about one-third of the people die.
- a If 100 shark attacks were reported in one year, how many would be Great White attacks?
 - b How many people would die from this number of Great White attacks?
- Great White Sharks are ovoviviparous – that is, the eggs are kept inside the female body until they hatch. About 40 eggs develop in the female’s uterus, hatching there and continuing to develop until they are born. The Great White has an 11 month gestation (pregnancy) period. There is not much room for the embryos to grow and they must eat each other to make more room – these sharks develop their predatory skills well before they enter the wild. On average, 8 Great White Sharks are born from an original litter of 40. Great White Sharks reach sexual maturity at about 12 years and live for 40 to 100 years.
- 9 What is the probability of a Great White being born?
- 10 Assuming the shark can get pregnant again straight after she gives birth, how many sharks could be born to a Great White female that died at the age of:
- a 50 years?
 - b 100 years?
- 11 Sharks are popular for their flesh – the ‘flake’ in ‘fish-n-chips’ is shark. About 2 190 tonnes of shark is landed from the Southern Shark Fishery each year. This makes the annual wholesale value of the fishery AU\$13.831 million. Each shark caught weighs about 15 kg.
- a How many sharks are caught per year in the Southern Park Fishery?
 - b What is the approximate wholesale monetary value of one of these sharks?



Whales are mammals and are more closely related to humans than to the many fish and creatures they share the oceans with. Like us, they are warm-blooded, give birth to live young and suckle their young. Baleen Whales are characterised by having baleen plates for filtering food from water, rather than teeth. This distinguishes them from the other group of whales, the toothed whales or Odontoceti.

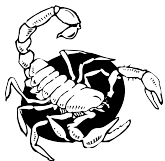
Whenever you see a colour code letter, find your answer in the shape at the end of the worksheet and colour that section. Colour code: R = red, B = blue, G = green, Y = yellow.

- 1 Find the meaning of each word listed below.
a weaning b lunge c breach
- 2 Whales are the largest living animals on Earth. The average length and weight of some species are shown in the table. In your workbook, draw column graphs to compare the length and weight of these whales with that of an elephant and an orca. Draw one graph for length and a second graph for weight.

Whale	Length (m)	Weight (t)
Southern Right*	18	100
Grey*	14	35
Minke*	8.5	10
Sperm	15	45
Humpback*	14	50
Blue*	33	150
Elephant	2.6	4
Orca (dolphin)	9.5	0.8
* Baleen whales, which have no teeth		

Commercial whaling in Australia started in about 1820. By 1845, twenty-six thousand Southern Right Whales had been killed. Almost 100 years later, they were a protected species. Between 1936 and 1938, approximately twelve thousand Humpback Whales were killed. A 10 year moratorium on Humpback whaling was declared in 1939. After killing commenced again in 1949, almost sixteen thousand Humpbacks were killed in Australia by 1963. Australia ceased whaling in 1978.

- 3 Calculate the average number of whales killed per year in Australia between:
a 1820 and 1845. (R) b 1936 and 1938. (B) c 1949 and 1963. (G)
- 4 Suggest an explanation for the large increase in your average answer to question 3b compared to 3a.
- 5 During the whaling operations, it was found that approximately 1 200 whales provided eleven thousand tonnes of oil.
a Calculate the average weight of oil produced from one whale, in kilograms. (Y)
b If a whale weighs 45 tonne, what percentage of the whale is oil? (R)
- 6 In 1963, aerial and surface surveys indicated that 300 to 500 Humpback Whales remained on the east coast of Australia. In 2005, approximately two thousand whales were observed in the same oceans. Humpbacks, along with the Bowhead, Blue and Right Whales, are some of the most endangered species on Earth.
a On flights down the east coast of Australia in 1963, one observer saw 300 whales and another saw 500. What was the average number of whales observed? (Y)
b Calculate the increase in the number of whales from the 1963 average to the 2005 figure of 2000. (Y)
c Calculate the average yearly increase in whale numbers over the 42 years from 1963 to 2005. (Y)
d Calculate the percentage increase in whale numbers from 1963 to 2005. (Y)



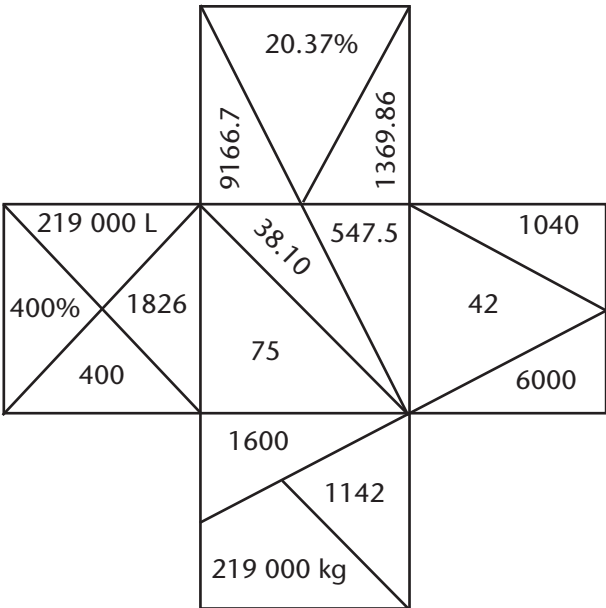
- 7 Humpback Whales breed every two years, normally giving birth to one calf. The newborn calf is approximately four metres long and weighs 1.5 tonne. The Humpback lives for fifty to one hundred years, but some have been known to live much longer. Female whales reach sexual maturity at about 7 years of age.
- a Three whales reached the ages of 50 years, 70 years and 105 years, respectively. Calculate their average lifespan. (Y)
 - b If a female whale lives for 91 years, calculate the number of whales she could produce in her lifetime. (R)
 - c Suggest why it takes so long to re-establish whale populations after extensive hunting has occurred.

- 8 A young Humpback Whale drinks about 600 litres of milk per day from its mother. After weaning, it eats approximately 500 tonnes of krill (a small shrimp-like creature about 5 cm in length) each year. An adult whale eats 1.5 tonne of food each day.
- a How many litres of milk are consumed by a calf in its first year? (Use 1 year = 365 days.) (Y)
 - b How many kilograms of milk is this? (Use 1 L weighs 1 kg.) (G)
 - c How many kilograms of krill does a young whale eat per day? (R)
 - d A large piece of steak from the butcher weighs about 0.75 kg. How many 0.75 kg pieces of steak would you need to equal the weight of krill eaten daily by a young whale? (Y)
 - e How many tonnes of food does an adult whale eat in a year? (R)

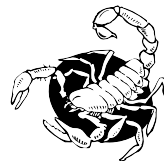
- 9 When a whale lunges, approximately forty percent of its body launches out of the water. When breaching, the whale turns and lands on its back eighty percent of the time.
- a Use the table in question 2 to calculate the length which would be out of the water when lunging, for each whale.
 - b If a whale breaches 45 times, how many times does it land on its back?

- 10 In most species of Baleen Whales, the females are about 10% heavier than the males. This situation is reversed for the larger toothed whales, where the males are larger. For example, Sperm Whale females and Orca females are 40% lighter than the males. Use the table in question 2 to calculate the average weight of the females, if the weights listed are for the males.

- 11 Cut out the 4 shapes you coloured in the diagram. Rearrange them to make a square.



Mighty molluscs - octopus, squid & cuttlefish



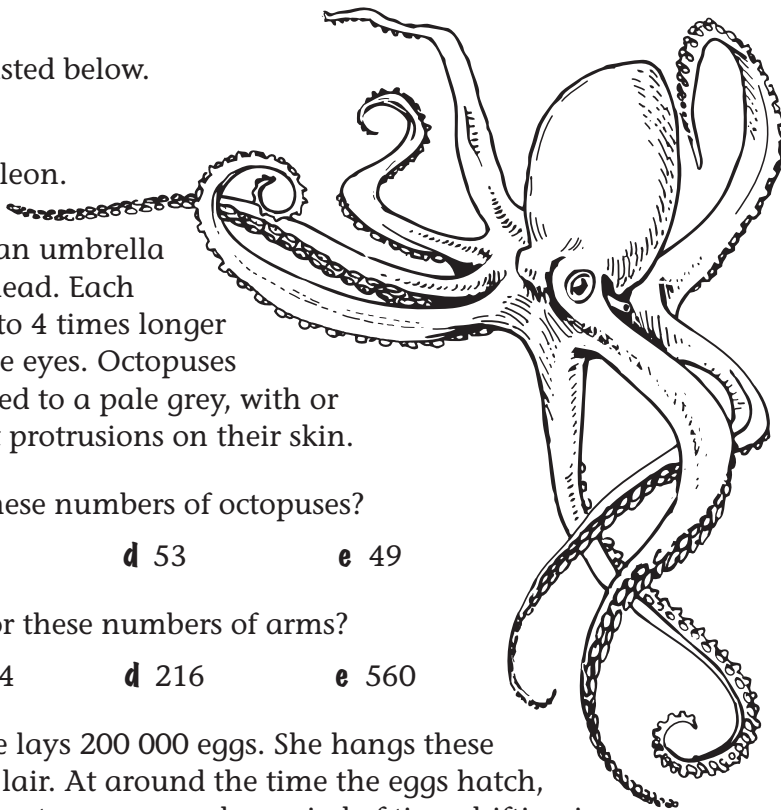
4

Octopus, squid and cuttlefish live in the oceans and do not have backbones. They are Cephalopoda, which means 'head foot'. Members of this group have feet or tentacles connected to the head, not the body. They can change the colour of their skin to mimic their environment and hide from predators. They also eject a thick blackish ink in a large cloud to help them escape from predators. They can move quickly by sucking water into their mantle cavity and quickly squirting it out of their siphon in a fast, strong jet.

1 Find the meaning of each word listed below.

- | | |
|---------------|--------------|
| a lair | b larval |
| c respiratory | d chameleon. |

Octopuses have 8 arms which create an umbrella shape around their central rounded head. Each arm has two rows of suckers and is 3 to 4 times longer than the body. The head has two large eyes. Octopuses can be many different colours, from red to a pale grey, with or without mottling and with or without protrusions on their skin.



2 How many arms altogether on these numbers of octopuses?

- | | | | | |
|-----|-----|------|------|------|
| a 5 | b 7 | c 25 | d 53 | e 49 |
|-----|-----|------|------|------|

3 How many octopuses are there for these numbers of arms?

- | | | | | |
|------|------|-------|-------|-------|
| a 16 | b 24 | c 104 | d 216 | e 560 |
|------|------|-------|-------|-------|

In some species of octopus, the female lays 200 000 eggs. She hangs these eggs in strings from the ceiling of her lair. At around the time the eggs hatch, the mother dies and the young larval octopuses spend a period of time drifting in clouds of plankton where they feed.

4 Calculate the total number of eggs these numbers of female octopuses would lay.

- | | | | | |
|-----|-----|-----|------|------|
| a 5 | b 7 | c 9 | d 15 | e 24 |
|-----|-----|-----|------|------|

5 How many octopuses are needed to lay these quantities of eggs?

- | | | | | |
|-----------|-----------|-----------|-------------|-------------|
| a 400 000 | b 600 000 | c 100 000 | d 1 000 000 | e 5 000 000 |
|-----------|-----------|-----------|-------------|-------------|

6 The Blue Ringed Octopus carries enough poison to kill 26 human adults within minutes. How many human adults could these numbers of Blue Ringed Octopuses kill?

- | | | | | |
|-----|-----|------|------|------|
| a 6 | b 8 | c 15 | d 21 | e 34 |
|-----|-----|------|------|------|

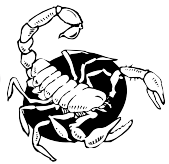
7 How many Blue Ringed Octopuses are needed to kill these numbers of human adults?

- | | | | | |
|------|------|------|-------|------|
| a 26 | b 52 | c 78 | d 104 | e 69 |
|------|------|------|-------|------|

When Blue Ringed Octopus mate in early autumn, the male dies. The female lays approximately 50 eggs in late autumn. Once the eggs are hatched 6 months later, the female dies. The young octopuses develop quickly, and mature and mate the following autumn.

8 Draw a timeline to show the life cycle of the Blue Ringed Octopus.





9 When bitten by a Blue Ringed Octopus, a feeling of sickness is felt within a few seconds. Within 10 seconds, the victim is blind. Within one minute, there is a loss of sense of touch, speech and the ability to swallow. Paralysis sets in and the body goes into respiratory arrest within a further 3 minutes. Draw a timeline to show the effects of the octopus bite.



10 When referring to octopus flesh, fishers use terms such as 'U/25'. This is pronounced 'under twenty-five' and refers to the number of octopuses per kilogram – in this case, fewer than twenty-five octopuses per kilogram. 26!!40 is pronounced 'twenty-six forty' and means that there are twenty-six to forty octopuses per kilogram. For each weight listed, decide if the octopus would be classified as U/25 or 26!!40.

- a 35 g b 50 g c 60 g d 25 g e 1.2 kg

11 The table shows the live weight of octopus caught in two managed fisheries.

a Complete the total column of the table.

b Use the data in the table to draw a line graph for the total catch.

c Comment on any trends.

Year	Cockburn Sound (kg)	Western Rock (kg)	Total (kg)
1995	280	114 000	
1996	1 900	136 000	
1997	3 400	156 000	
1998	5 000	165 000	
1999	2 870	122 000	
2000	2 200	154 000	
2001	9 100	181 000	
2002	8 600	211 000	
2003	28 000	185 000	
2004	21 000	178 000	

12 A bus is 8 m long, a Sperm Whale is 15 m long, a Giant Squid is 13 m, a Colossal Squid (largest eyes in the animal kingdom) is 20 m, a Calamari Squid is 60 cm, and an Arrow Squid is 75 cm. In your workbook, draw a horizontal bar graph to compare these lengths.

13 80% of the weight of a squid can be used for eating purposes. Calculate the weight in grams of flesh for eating that could be recovered from squid with the weights listed below.

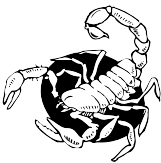
- a 300 g b 500 g c 600 g d 1.5 kg e 2.4 kg

14 Listed below are the weights of flesh recovered from 5 different squid. Calculate the original weight of each squid, to the nearest gram.

- a 350 g b 260 g c 1200 g d 1.4 kg e 1.1 kg

15 Cuttlefish are sometimes called the chameleons of the sea because of their remarkable ability to rapidly alter their skin colour. Their skin changes very quickly to camouflage them from predators. The colour change is produced by groups of red, yellow, brown and black pigmented cells. If there are 200 of these specialised pigment cells per square millimetre of skin, how many are there on these areas of skin?

- a 5 mm² b 12 mm² c 15 mm² d 1 cm² e 5.1 cm²



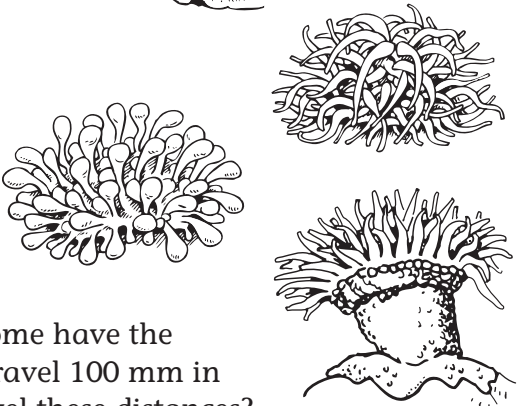
Sea anemones are a group of water-dwelling, predatory animals. They are named after the anemone, a flower found in many gardens. Sea anemones are closely related to coral and jellyfish.

A jellyfish has no head, brain, heart, eyes or ears. It has no bones, either. To capture prey for food, jellyfish have a net of tentacles that contains poisonous, stinging cells. When the tentacles brush against prey – or a person’s leg – thousands of tiny stinging cells explode, launching barbed stingers and poison into the victim.

Whenever you see a colour code letter, find your answer in the shape at the end of the worksheet and colour that section. Colour code: R = red, B = blue, G = green, Y = yellow.

- 1 Find the meaning of each word listed below.
a transparent b nauseous c knot (as in water speed) d tentacle
- 2 Draw bar graphs of the data in the table to compare the size and tentacle length of anemones to a large coffee mug.

Name	Diameter (mm)	Tentacle length (mm)
a Coffee mug	60	Height 100
b Adhesive Sea Anemone	300	5
c Bulb-Tentacle Sea Anemone	50	100
d Beaded Sea Anemone	200	50
e Leathery Sea Anemone	500	100
f Magnificent Sea Anemone	1 000	75
g Mertens’ Sea Anemone	1 200	40

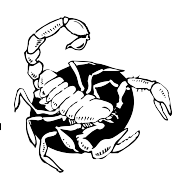


- 3 Most sea anemones spend their lives in one place. Some have the ability to move, but if they do move, they can only travel 100 mm in an hour. How long would it take an anemone to travel these distances?
a 200 mm b 400 mm c 650 mm d 1 m e 2.6 m
- 4 An individual anemone can glide on its pedal disc, covering about five millimetres in a day. Sometimes it might detach entirely, and roll or be carried by the waves.
a How many millimetres an hour does this anemone move?
b Change this speed to kilometres per hour.

- 5 Draw a picture graph to compare the relative sizes of the jellyfish listed in the table.

Jellyfish name	Diameter
Irukanji	2 cm
Sea Nettle	30 cm
Box Jellyfish	20 cm
Portugese Man-Of-War	30 cm
Nomura	2.1 m
Blue Jellyfish	2 m

Box Jellyfish are pale blue and transparent, so they are difficult to see even in clear ocean waters. For many years, ocean swimmers sometimes suffered from mysterious, excruciating pain, often followed by death within two to three minutes. Scientists eventually discovered that this was caused by the Box Jellyfish. Measuring up to 20 cm along each side of its cube or bell, the Box Jellyfish has up to 15 tentacles on each corner, which can be 3 metres in length with up to 5 000 nematocysts (stinging cells) on each one.



6 Draw a scale picture of the Box Jellyfish, to a scale of 1:20.

The Irukandji Jellyfish is a small jellyfish with a bell (head) approximately 2 cm in diameter. The initial sting of the jellyfish is usually not very painful. About 30 minutes after being stung, the victim develops a severe backache or headache, and shooting pains in their muscles, chest and abdomen. They may also feel nauseous, anxious and restless.

In one season, approximately 120 patients were treated at a local hospital in Cairns for 'Irukandji syndrome'.

7 Out of the 120 patients, calculate the number of people in each of the following categories.

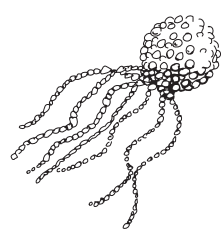
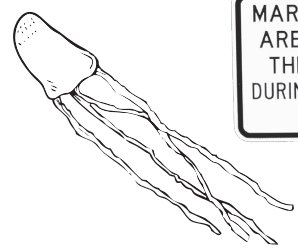
- a 20% occurred on a reef. (R)
- b 22% were tourists. (B)
- c 70% were local residents. (G)
- d 25% of the stings occurred on the hands, face or feet. (Y)
- e 45% had no first aid treatment before arrival at the hospital. (R)

8 Out of the 120 patients, calculate the percentage of people in each of these categories.

- a 23 people had heart problems. (G)
- b 5 people had demonstrable heart problems. (Y)
- c 2 people had severe heart problems. (G)
- d 44 people were discharged within 6 hours. (R)
- e 46 people were discharged after 24 hours. (B)



MARINE STINGERS ARE PRESENT IN THESE WATERS DURING THE SUMMER MONTHS



Jellyfish feed on the same kinds of prey as fish. If fish are removed from an environment, jellyfish are likely to move in. Depleted fish stocks have decreased competition for jellyfish, allowing them to breed to plague proportions. Japan has experienced plagues of the giant jellyfish Nomura, which can grow to more than a metre wide and weigh up to 150 kg. These jellyfish cause major problems if they become caught in fishing gear, boat engines or mechanical equipment. Fishers have reported cases of more than 800 of these jellyfish being caught in a single net.

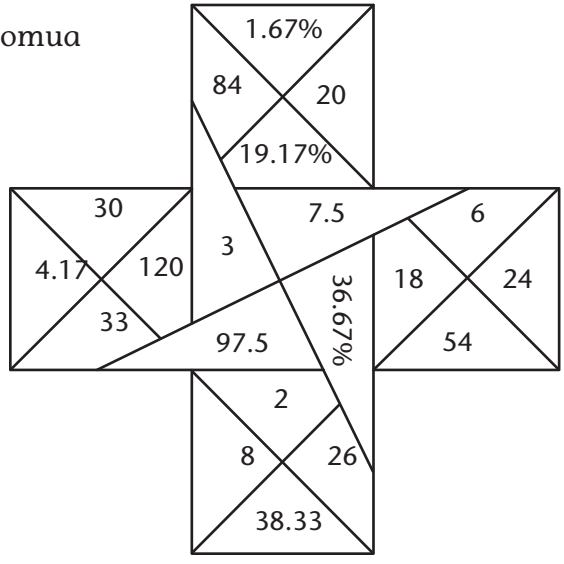
9 Calculate the total weight (in tonnes) of 150 kg Nomura Jellyfish, if these numbers were caught in a net.

- a 40 (R) b 50 (G) c 800 (Y)
- d 65 (B) e 220 (Y)

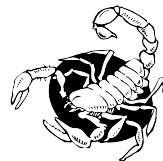
10 How many 150 kg Nomura Jellyfish would weigh these quantities?

- a 300 kg (B) b 450 kg (Y)
- c 1 200 kg (B) d 3 tonne (G)
- e 2.7 tonne (R)

11 Cut out the shapes you coloured in the diagram. Rearrange them to make a square.



Marinara on the menu - oysters and scallops



6

Bivalves are molluscs. They have two-part shells. This group of animals includes scallops, clams, oysters and mussels. Bivalves are exclusively aquatic and are found in marine and fresh water.

- 1 Find the meaning of each word listed below.

a salinity

b hessian

The oyster becomes sexually mature at about 1 year of age, and then always as a male. After about 3 years, it can change sex and reproduce as a female. From then on it can change sex back and forth throughout its life. When reproduction takes place, the male releases its sperm into the water and the female sucks it in to her shell and fertilises the eggs. A female can contain as many as 1.1 million eggs. Fertilised eggs develop within the female in the first 10 days. The larvae swim freely in the water for a couple of weeks and so spread to other areas. Afterwards, the larvae fall to the bottom and land on rocks, shells or stakes. Then the oyster is known as a brood and it is about 0.3 mm in size. The oyster lives for about 8 years.

- 2 Draw a timeline on the line below to show the life cycle of an oyster. Then in your workbook draw and annotate a diagram of the life cycle.



- 3 If a female fertilises 1.1 million eggs, how many oysters would grow to adulthood for each of these survival rates?

a 1%

b 2%

c 0.002%

d 0.003%

e 0.0001%

- 4 If reproduction is to take place, salinity is important – there must be at least 1 part salt to 40 parts water. Simplify the salinity of these batches of water, and decide if reproduction can take place.

a 2:20

b 2:100

c 5:400

d 4:120

e 2:90

Oysters are filter feeders – that is, they consume a variety of species of phytoplankton, bacteria and viruses from the surrounding water. Adult oysters can filter 10 L of water per hour as they search for food.

- 5 How much water could an oyster filter in:

a 2 hr?

b 3 hr?

c 6.5 hr?

d 5 hr 30 min?

e 6 hr 25 min?

- 6 How long would it take an oyster to filter:

a 50 L?

b 70 L?

c 65 L?

d 125 L?

e 241 L?

The harvesting of oysters in oyster farms for market occurs prior to spawning, when the oysters are in peak condition. Oysters are brought in from the farm and graded into various shell sizes, such as bistro 50–60 mm, plate 60–70 mm and standard 70–85 mm. The oysters are then packed into hessian bags. For bistro oysters, 45 dozen oysters are packaged in each bag. Then the oysters are transported in refrigerated trucks directly to the wholesaler or retailer. Oysters stored at 5°C have a shelf-life of less than 10 days.

- 7 How many hessian bags are needed to store these numbers of bistro oysters?

a 90 dozen

b 225 dozen

c 440 dozen

d 4 860 oysters

e 24 300 oysters

8 Complete the table by deciding whether each delivery of oysters is safe to eat.

	Delivery date to store	To be consumed on	Safe to eat?
a	15th March	18th March	
b	20th May	29th May	
c	2nd April	15th April	
d	24th July	4th August	
e	28th April	10th May	

9 The world record for eating oysters is 288 oysters in 1 minute, 33 seconds.

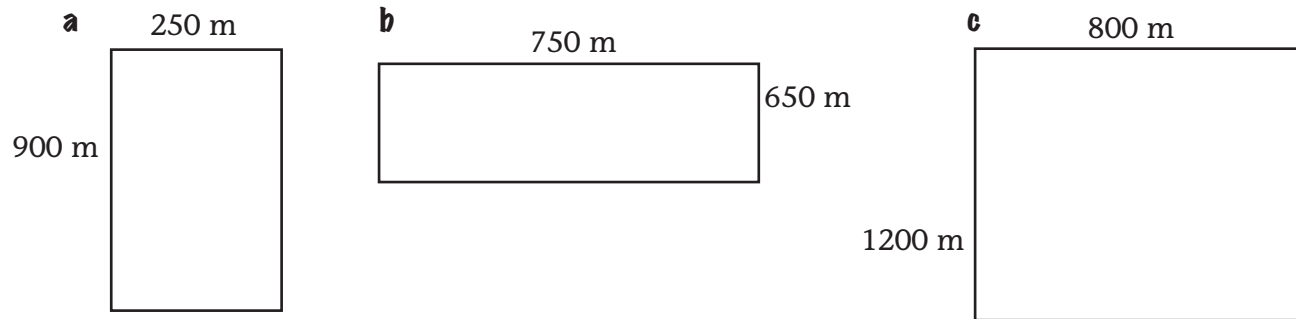
- a How many oysters is this per second?
- b How many dozen oysters is this?



Scallops are edible bivalves, similar to oysters and clams. They are found on the bottom of bay waters and in the sea. They do not attach themselves to a permanent anchorage. Instead, they move themselves through the water by opening and closing their shells. As a result, the muscle that controls the ‘hinge’ of the shell is much larger than that of oysters or clams.

Scallops in Bass Strait are collected in dredges which are towed along the seabed by vessels travelling at four to six knots. The dredges are made from strong, wire-mesh boxes that are four metres long, with a mouth about three metres wide and 30 cm high. 2.5 scallops per square metre is considered the minimum economic limit for dredge fishers.

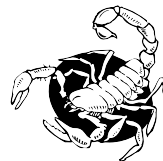
10 At the minimum economic limit, how many scallops would be obtained from these dredge areas?



11 Concentrations of 30 scallops per square metre were once common but over 4 years, this was reduced by starfish to five per square metre. For each dredge area in question 10, calculate the:

- a number of scallops dredged from the sites before the starfish infestation.
- b number of scallops dredged after the starfish infestation.
- c percentage reduction in the scallop catch.

12 Describe the effect of the starfish on scallop populations.

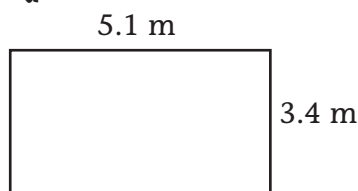


Seahorses belong to the family of fish which also includes pipefish and leafy sea dragons. There are over 32 species of seahorse. They are mainly found in tropical and subtropical coastal and reef waters all over the Pacific, Atlantic and Indian Oceans. All seahorses have bodies that are protected by strong external plates, arranged into a series of 'rings'. These rings help to protect the body of the fish, but also result in the fish having only limited flexibility of the body.

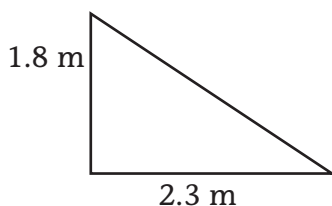
Seahorses stay in small areas. On average, males stay in one square metre of habitat and females in one hundred times that area. They bob around in sea grass meadows, mangrove stands and coral reefs where they are camouflaged by their murky brown and grey patterns which blend into the sea grass backgrounds.

- 1 How many male seahorses could be found in each of these habitats?

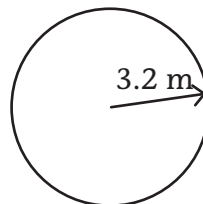
a



b



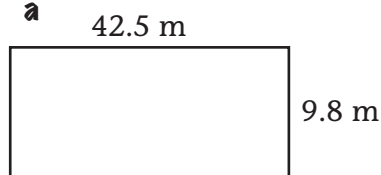
c



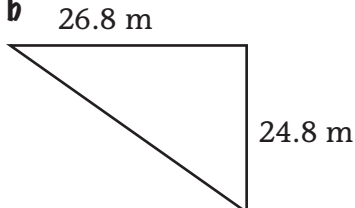
- 2 What is the area of the habitat of a female seahorse?

- 3 How many female seahorses could be found in these habitats?

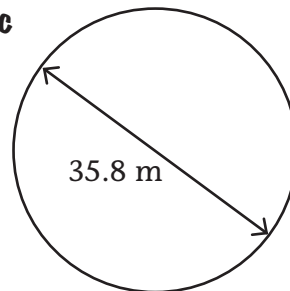
a



b



c



- 4 Seahorses range in adult head-to-tail sizes from 13 mm to 35 cm.
Draw a scale on the line below and mark in the range of seahorse lengths.



Seahorses and pipefishes are notable for being the only species in which males become 'pregnant'. The male seahorse can give birth to 2 000 babies at a time. Pregnancies last anywhere from 10 to 25 days.

- 5 How many babies could these numbers of males give birth to?

a 5

b 6

c 12

d 15

e 23

- 6 How many males are needed to give birth to these numbers of young seahorses?

a 24 000

b 26 000

c 28 000

d 13 000

e 9 000

Out of every 1000 infants born, fewer than five survive to adulthood, which helps to explain why litters are so large. The survival rates of these infants is actually fairly high compared to most fish. This is because they are sheltered in their father's pouch during the earliest stages of development. The eggs of most other fish are abandoned immediately after fertilisation.



- 7 What percentage of young seahorses survive to adulthood?
 - 8 How many seahorses would reach adulthood from a litter of:

a 1 600?	b 2 200?	c 300?	d 3 000?	e 4 200?
----------	----------	--------	----------	----------
- Seahorse populations have been endangered in recent years by overfishing. In Australia, seahorse fishing is strictly illegal. The seahorse is used in traditional Chinese herbology, and as many as 20 million seahorses might be caught each year and sold for this purpose. Unfortunately, seahorses around the world are rapidly moving towards becoming an endangered species. There is a huge trade in seahorses and their underwater habitats are threatened by pollution.
- 9 At an annual rate of 20 million, how many seahorses on average are caught in:

a 1 day?	b 6 days?	c 1 month?	d 3 weeks?	e 4 weeks?
----------	-----------	------------	------------	------------
 - 10 On average, how long would it take to catch these quantities of seahorses, to the nearest day?

a 42 million	b 500 000	c 330 000	d 16 million	e 25 million
--------------	-----------	-----------	--------------	--------------

Seahorses lack a tail fin. Since this fin provides most fish with their main swimming power, seahorses don't swim very fast. However, they are extremely maneuverable, and can hover precisely. Seahorses can swim forwards or backwards by moving their dorsal and pectoral fins. The main thrust comes from their dorsal fin, which beats 70 times a minute. Pectoral fins on each side of the face are used to steer.

- 11 How many times does the dorsal fin beat in:

a 3 minutes?	b 5 minutes?	c 10 seconds?	d 3 hours?	e 1.2 hours?
--------------	--------------	---------------	------------	--------------
- 12 Calculate the beat rate of the dorsal fin of these seahorses, in beats per minute.

a 490 beats in 5 minutes	b 210 beats in 3 minutes
c 450 beats in 6 minutes	d 600 beats in 8.1 minutes
e 2 beats in 0.8 seconds	

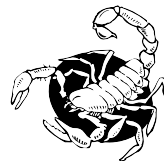
A seahorse can consume 3 000 brine shrimp each day. Seahorses have no teeth and swallow their food whole.

- 13 How many brine shrimp could a seahorse eat in:

a 5 days?	b 21 days?
c 4 weeks?	d 2 months?
e 2 hours?	
- 14 How long would it take a seahorse to eat these numbers of brine shrimp?

a 7 000	b 10 000
c 8 000	d 25 000
e 34 000	





Starfish or sea stars are any echinoderms belonging to the class Asteroidea. They typically have five or more 'arms' which radiate from a central disk in a symmetrical way. Most starfish have 5 arms, but some have more or fewer.

1 Find the meaning of each word listed below.

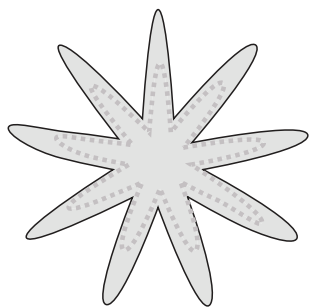
a radiate

b regenerate

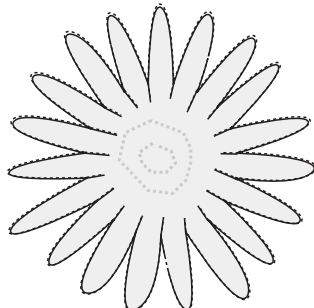
c spawning

d buoyant

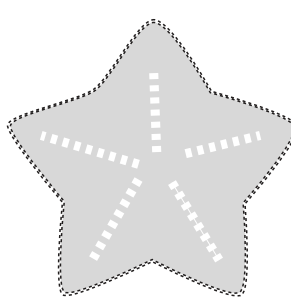
While starfish are commonly represented as having only 5 legs, there are some that exhibit many more, as you can see below.



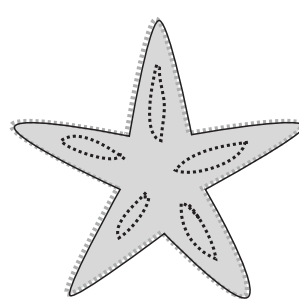
Nine Armed Starfish



Sunflower Starfish



Cushion Starfish



Spiny Beaded Starfish

2 On each starfish shown above, draw a line to mark one axis of symmetry. Then draw at least two more lines to mark further axes of symmetry.

3 How many legs on:

a 40 Nine Armed Starfish?

b 34 Cushion Starfish?

c 234 Spiny Beaded Starfish?

d 321 Sunflower Starfish?

e 45 Cushion Starfish?

f 567 Sunflower Starfish?

4 How many starfish are needed for each of these groups of legs?

a 243 Nine Armed Starfish legs

b 1 197 Sunflower Starfish legs

c 2 050 Cushion Starfish legs

d 2 600 Spiny Beaded Starfish legs

e 7 020 Nine Armed Starfish legs

f 10 640 Sunflower Starfish legs

5 The underside of a starfish is covered with hundreds of tube feet. These are used for walking around, for attaching tightly to rocks and for holding on to prey. With so many little legs, some starfish can reach speeds of 120 centimetres per minute.

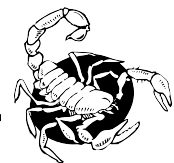
a How many centimetres per hour is this?

b How many metres per hour is this?

c How many kilometres per hour is this?

6 The Crown of Thorns Starfish can move up to 600 m in a week. How fast on average is this, in metres per hour?

Some species of starfish have the ability to regenerate lost arms and, over time, can grow an entire new arm. Most species must have the central part of the body intact in order to regenerate, but a few can grow an entire starfish from a single ray.



- 7 Calculate the percentage of starfish that grew new arms from the following batches examined by marine scientists.
- | | | |
|-------------------|-------------------|-------------------|
| a 203 out of 1900 | b 122 out of 1234 | c 163 out of 1825 |
| d 211 out of 2098 | e 102 out of 1291 | f 198 out of 2023 |

The Crown of Thorns Starfish has thorn-like spines sprouting all over its body for protection. These starfish grow to a diameter of more than 50 cm and have 12 to 20 arms extending from their centre. They are the largest starfish in the world, after the Giant Sunstar. The Crown of Thorns Starfish is a coral reef predator and preys on coral polyps. It is a serious threat to the coral on the Great Barrier Reef and on other coral reefs around the world.

- 8 It takes a Crown of Thorns Starfish about 5 hours to eat a small area of coral, about the size of a small plate.
- a If a small plate measures 6 cm in diameter, calculate the area of coral a Crown of Thorns Starfish can eat in one year.

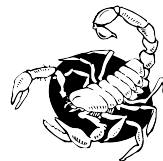


- b How much is this to the nearest square metre?
- c Calculate the side length of a square this size.
- d Use chalk to mark out the square on some flat concrete.
- 9 The Crown of Thorns Starfish can release 20 million eggs in a summer spawning. Many of the eggs are lost due to poor fertilisation, predators and a lack of food. How many extra adult starfish are there, if for 20 million eggs there is a success rate of:
- | | |
|------------------------------|-----------------------------|
| a 0.001% reaching adulthood? | b 0.01% reaching adulthood? |
| c 25% reaching adulthood? | |

The eggs of the Crown of Thorns Starfish (COT) are not buoyant and drop to the ocean floor. After 2 days, the starfish is 0.05 mm, after 6 months it is 20 mm, after 2 years it is 14 cm (and mature) and after 3 to 4 years it is about 35 cm in size and its growth declines dramatically. It is thought that these starfish can live for up to 8 years.

- 10 Draw a time versus size graph on the axis at the right to show the growth of the Crown of Thorns Starfish over its first 5 years. (Use days on the horizontal axis.)
- 11 Use your graph to answer these questions.
- | |
|--|
| a What is the diameter of a COT when it is 1 000 days old? |
| b What is the diameter of a COT when it is 500 days old? |
| c How old is a COT at 200 mm diameter? |
| d How old is a COT at 300 mm diameter? |





The Golden Perch, commonly called a yellow belly, is a medium-to-large fish with a deep, laterally compressed body. It is considered rare in much of its original range, although it is still present in large numbers in many locations. Its maximum weight is 23 kg and maximum length is 760 mm, but it is usually less than 400 mm and 4 kg. The body colour is generally olive-green with a yellow or cream belly. The mouth is large, with the lower jaw protruding slightly. The tail is rounded.

Murray Cod are an Australian native fish. They were once widespread throughout the Murray-Darling system, and found in some surprisingly small streams. This species was the most common of the larger native fish in the Murray-Darling system at the time of European settlement. Wild stocks are now estimated to be less than ten percent of the population present at that time.

Whenever you see a colour code letter, find your answer in the shape at the end of the worksheet and colour that section. Colour code: R = red, B = blue, G = green, Y = yellow.

1 Change these Golden Perch measurements to the units shown.

- | | | |
|----------------------------------|---------------------------------------|---------------------------------|
| a 3 kg = ____ g (Y) | b 9 kg = ____ g (B) | c 9.8 kg = ____ g (B) |
| d 8 kg 450 g = ____ g (G) | e 5 000 g = ____ kg (R) | f 7 000 g = ____ kg (G) |
| g 6 500 g = ____ kg (G) | h 9 450 g = ____ kg ____ g (Y) | i 700 mm = ____ cm (Y) |
| j 760 mm = ____ cm (B) | k 450 mm = ____ cm (R) | l 5 600 mm = ____ cm (B) |
| m 50 cm = ____ mm (R) | n 500 cm = ____ mm (B) | o 650 cm = ____ mm (G) |
| p 4 560 mm = ____ cm (Y) | | |

Golden Perch are predominantly found in warm, turbid, slow-flowing rivers. The species is long-lived, with a maximum age of about 25 years, although most live less than 11 years. Males are reproductively mature at 2 years and females at 4 years.

2 Change these Golden Perch ages to the units shown.

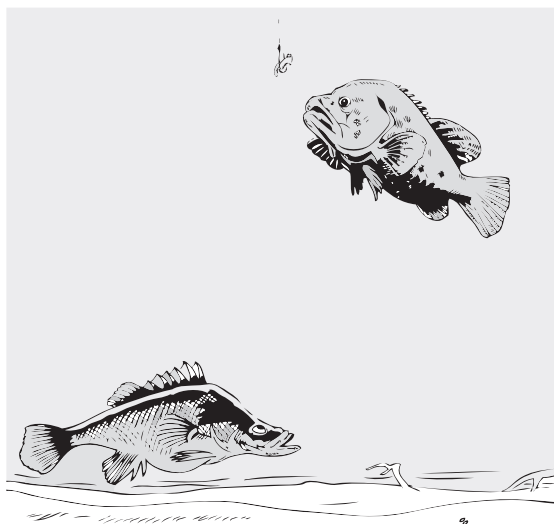
- | | | |
|-----------------------------------|---------------------------------|-----------------------------------|
| a 25 yr = ____ wk (R) | b 11 yr = ____ wk (G) | c 2 yr 23 wk = ____ wk (Y) |
| d 4 yr 12 wk = ____ wk (G) | e 3 wk = ____ d (R) | f 51 wk = ____ d (B) |
| g 23 wk 5 d = ____ d (Y) | h 12 wk 6 d = ____ d (R) | |

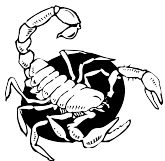
Adult and immature fish are migratory. Extensive upstream movements of more than 1 000 kilometres have been recorded for some adult fish. Outside the breeding season, individuals occupy home ranges of about 100 m² for weeks or months before relocating to another site where a new home range is established.

3 Change these Golden Perch movements to the units shown.

- | | |
|--------------------------|----------------------------|
| a 500 km = ____ m | b 600 km = ____ m |
| c 450 km = ____ m | d 1 000 km = ____ m |

4 Recent research in the Murray River suggests that some fish move downstream to spawn. Golden Perch are highly fecund, with a 2.3 kg female holding up to 500 000 eggs. Water-hardened eggs are large, about 3 mm diameter, semi-buoyant and drift downstream. Hatching occurs after 1 to 2 days and newly hatched larvae are about 3.5 mm long.





- a If the Golden Perch eggs were laid side-by-side in a straight line, how far in metres would they reach?
- b If all the eggs hatch and the larvae are laid end-to-end, how far in metres would they reach?

Murray Cod reach at least 60 cm and 3 to 4 kilograms in smaller streams. In larger rivers they grow to about 90 cm and weigh 20 kg. The largest Murray Cod ever officially recorded was 183 cm and 113 kg. Unfortunately, excessive fishing has led to a serious lack of large, mature breeding cod in wild populations. There are now very few cod above the minimum legal angling size of 50 cm. Egg counts range from less than 10 000 eggs for a barely mature female to approximately 90 000 for older females, which weigh around 22 kg.

- 5 How many eggs would be laid by these numbers of recently matured female cod?
- a 8 cod b 9 cod c 12 cod d 20 cod e 27 cod
- 6 How many eggs would be laid by these numbers of older female cod?
- a 7 cod b 6 cod c 13 cod d 21 cod e 41 cod

The eggs of cod are large, about 3.5 mm in diameter, and sticky. They are usually deposited onto hard surfaces such as logs, rocks or clay banks.

The male guards the eggs during incubation and they hatch after 5 to 13 days. The larvae are about 6 mm long at hatching and have a large yolk sac. Larvae drift downstream for 5 to 7 days, particularly by night in spring and summer.

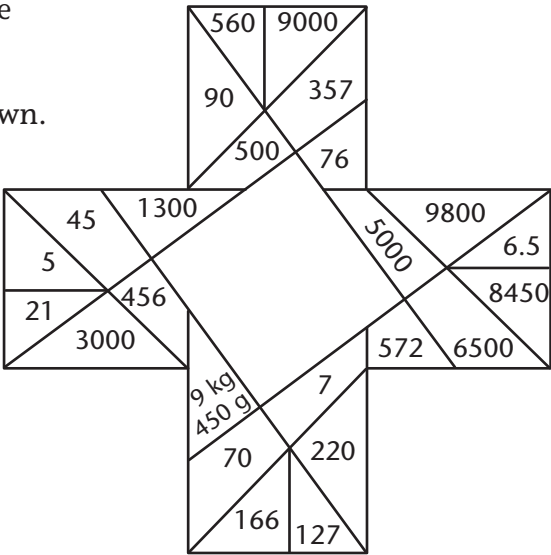
- 7 Change these cod life cycle times to the units shown.
- a 5 d = ____ hr b 13 d = ____ hr c 9 d 3 hr = ____ hr
- d 8 d 14 hr = ____ hr e 24 hr = ____ d f 72 hr = ____ d
- g 36 hr = ____ d h 80 hr = ____ d ____ hr

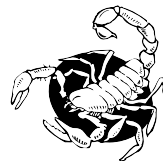
For a long time, fish researchers believed that Murray Cod lived near a particular snag. Large Murray Cod were fitted with transmitting radio tags, revealing that they make strong upstream spawning migrations in late winter and spring. Many fish travelled 40 to 50 kilometres upstream. Some travelled 120 kilometres upstream.

Most Murray Cod returned downstream to the same snag after spawning migrations. This remarkable homing behaviour is almost unknown in freshwater fish and emphasises the importance of snags to Murray Cod.

- 8 Change these cod travel distances to the units shown.
- a 4 km = ____ m b 50 km = ____ m
- c 120 km = ____ m d 45.7 km = ____ m
- e 4 000 m = ____ km f 30 000 m = ____ km
- g 3 400 m = ____ km h 12 300 m = ____ km

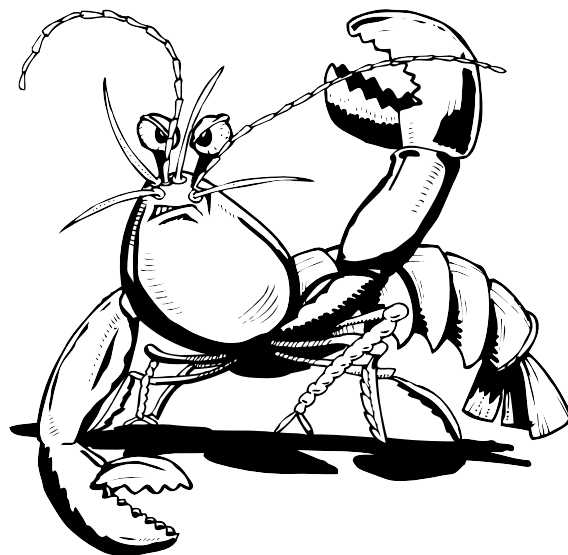
- 9 Cut out the 4 shapes you coloured in the diagram. Rearrange them to make a square.





*The common yabby, **Cherax Destructor**, is an Australian freshwater crustacean. In the wild, yabby populations remain strong. They have moved into new habitats created by reservoirs and farm dams. Their colour depends on the water clarity and habitat they live in – yabbies can range from black, blue-black or dark brown in clear waters to light brown or green-brown in 'dirty' waters.*

Yabbies occasionally reach up to 30 cm in length and up to 310 g in weight, but are more commonly 10 to 20 cm long. Like all crustaceans, yabbies have to moult as they grow. As they grow older, they moult less. After each moulting, their weight can increase by up to 50%.



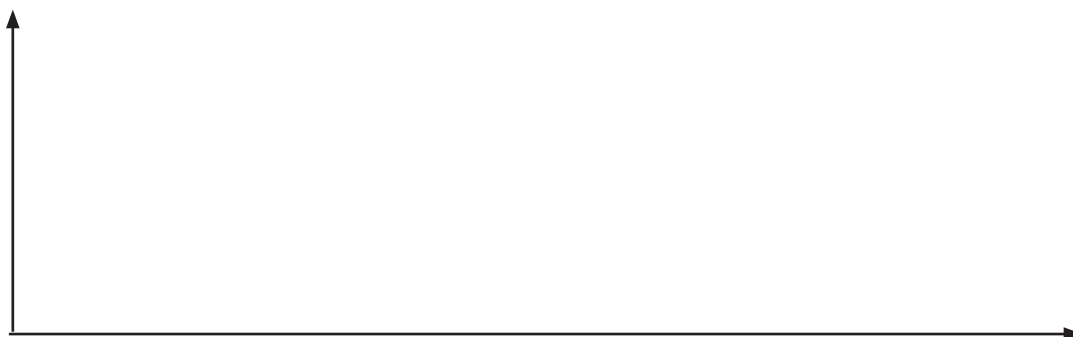
- 1 Calculate the new weight of each yabby if its weight increases by 40% after each moult.
a 20 g b 35 g c 25 g
d 58 g e 18 g

The number of fertilised eggs carried by the female yabby ranges from 100 to 300 for a young female to 1 100 for a larger, older female. The eggs are located under their tails.

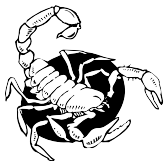
- 2 How many eggs could be carried by these numbers of older females?
a 5 yabbies b 10 yabbies c 15 yabbies d 12 yabbies e 20 yabbies
- 3 How many eggs could be carried by these numbers of young females, if each one carries 200 eggs?
a 5 yabbies b 10 yabbies c 15 yabbies d 12 yabbies e 20 yabbies

Juvenile yabbies are approximately 0.02 g upon hatching. The young yabby grows rapidly, gaining about 0.8 g in the first 60 days. The juvenile moults several times and continues to grow rapidly until it reaches adulthood and 80 grams after about 12 months. The maximum size of around 310 grams for adult yabbies is reached in 2 to 3 years. The minimum marketable size for yabbies is 30 grams, and this size can be reached in around 6 months.

- 4 Draw a line graph to show the growth rate of the yabby. Mark in the marketable size.



- 5 Use your graph to answer these questions.
a How heavy is a yabby aged 30 months?
b How old is a yabby that weighs 200 grams?

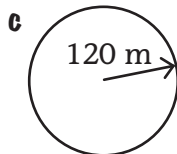
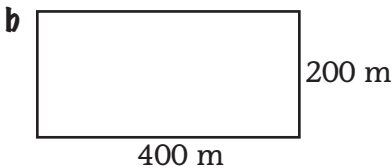
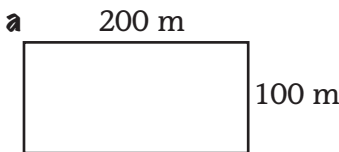


Yabbies are commonly found on muddy or silted bottoms of dams and rivers. They are rarely found in clear water habitats, preferring muddy water. Muddy waters afford some protection from predators such as fish and birds, giving the yabby a better chance of survival. Optimal growth is achieved when water salinity (salt) levels are lower than 1 part salt to 125 parts water and the water is well oxygenated.

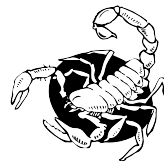
- 6 Calculate the ratio of salt to water (salinity level) of these samples of water.
- a 16 parts of salt and 4 000 parts of water
 - b 20 parts of salt and 20 000 parts of water
 - c 2 parts of salt and 200 parts of water
 - d 30 parts of salt and 27 000 parts of water
- 7 Yabbies are capable of surviving in sea water for approximately 48 hours. However, high salinity levels result in increased stress and the yabby dies. In which samples of water in question 6 are the yabbies likely to survive?
- 8 The tail meat of a yabby is a popular food for many people. This portion makes up about 15% of the total body weight when the yabby has been headed and shelled. Calculate the weight of tail meat that could be obtained from yabbies weighing:
- a 40 g.
 - b 80 g.
 - c 150 g.
 - d 275 g.
 - e 300 g.
- 9 Some of the costs associated with yabby farming are shown in the table. Fill in the missing values at a, b and c.

Set-up costs		Running costs		Income	Profit
Collecting trays	\$3 100	Feed	\$2 100	Sales: \$21 000	c
Nets	\$800	Fuel	\$2 500		
Cleaning buckets	\$50	Bait	\$230		
Grading tray	\$75	Ice	\$400		
Total:	a	Total:	b		

- 10 If the yabby farmer worked for 230 hours to gain the profit you calculated in the previous question, what is her hourly rate of pay?
- 11 Farmers producing yabbies from their dams have obtained yields of up to 650 kg of yabbies per hectare of water. Calculate the expected yield from dams with these quantities of water:
- a 3 ha
 - b 5 ha
 - c 12 ha
 - d 3.2 ha
 - e 2.1 ha
- 12 As yabby farmers adopt new technology and techniques, it is hoped that commercial yields in dams will reach 1.5–2.0 tonnes/ha/year. What would be the expected yield from dams of these sizes, at 1.75 tonnes/ha/year?
- a 4 ha
 - b 6 ha
 - c 11 ha
 - d 15 ha
 - e 5.6 ha
- 13 Calculate the annual yield at 1.75 tonnes/ha from dams of these sizes. (1 ha = 10 000 m²)

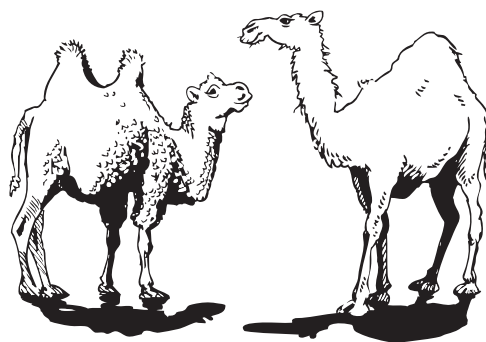
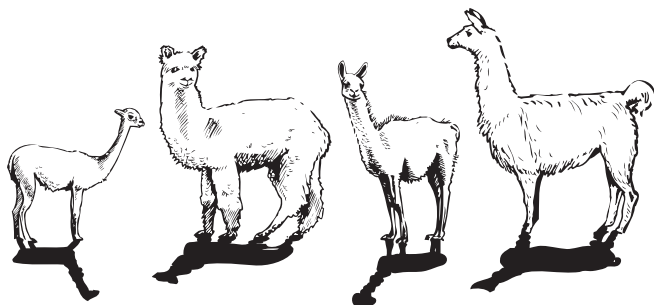


The Camelidae family - camels



11

There are two main types of camels – the Dromedary which is one-humped and the Bactrian Camel which has two humps. The term 'camel' is also used to describe any of the six camel-like creatures in the family camelidae. These are the two true camels and the four South American camelids:



1 Find the meaning of each word listed below.

a feral

b dehydrated

c hydrated

d ingest

A fully-grown adult camel stands 1.85 m at the shoulder and 2.15 m at the hump. The hump rises about 60 cm from its body.

2 Change these camel measurements to the units shown.

a 2 m = ____ mm

b 3 m = ____ mm

c 1.85 m = ____ mm

d 2.15 m = ____ mm

e 4 m = ____ cm

f 3 m = ____ cm

g 1.85 m = ____ cm

h 2.15 m = ____ cm

3 Currently there is a large feral population of Dromedary Camels. This population is estimated at 700 000 in central Australia. These feral camels are descended from the camels that were introduced as a means of transport in the 19th century. The feral population is growing at approximately 11% per year. If growth rates remain constant, how many camels will exist in Australia:

a after 1 year?

b after 2 years?

c after 4 years?

4 Camels are scattered throughout the arid interior of Australia, with an estimated 50% in Western Australia, 24% in the Northern Territory and 25% in western Queensland and northern South Australia. If there are 700 000 camels in total, how many are located in:

a Western Australia?

b the Northern Territory?

c all other parts of Australia?

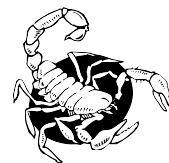
Camels can withstand changes in body temperature and water content that would kill most other animals. Their body temperature can range from 34°C at night to 41°C during the day. Only above this threshold will they begin to sweat. The safe body temperature range for a human is 35.6°C to 37.2°C.

5 Write a temperature scale on the arrow below. Then shade the temperature ranges for a human and a camel.



Camels can withstand a weight loss due to sweating of approximately 25%. Most mammals can only withstand 3.5% dehydration before heart failure results from the thickened blood. A camel's blood remains hydrated, even though the body fluids are lost, until this 25% limit is reached.

The Camelidae family - camels (cont'd)



11

- 6** Calculate the maximum safe weight loss for a camel of each weight listed.
a 300 kg **b** 550 kg **c** 220 kg **d** 350 kg **e** 490 kg
- 7** Calculate the maximum safe weight loss in these other mammals if they weigh:
a 85 kg **b** 90 kg **c** 210 kg **d** 350 kg **e** 490 kg

Camels eating green herbage can ingest sufficient moisture in milder conditions to maintain their body's hydrated state without needing to drink. In extremely dry areas, they need access to water holes. A thirsty camel can drink 200 litres in three minutes. It can drink water that is too salty or brackish for other animals to drink.

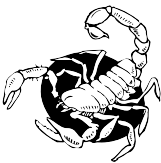
- 8** How much water can a camel drink in:
a 6 min? **b** 9 min? **c** 1 min? **d** 5 min? **e** 2 min 30 s?
- 9** How long would it take a camel to drink:
a 400 L? **b** 800 L? **c** 100 L? **d** 120 L? **e** 250 L?

A camel carcass can provide a substantial amount of meat. The male Dromedary carcass can weigh 400 kg or more, while the carcass of a male Bactrian can weigh up to 650 kg. The carcass of a female camel weighs less than the male, at about 330 kg. The hump is considered to be a delicacy.

- 10** How much meat could be obtained from these numbers of male camels, in tonnes?
a 5 Dromedary Camels **b** 7 Bactrian Camels
c 12 Bactrian Camels **d** 9 Dromedary Camels
- 11** How many female camels are needed to supply these quantities of meat?
a 660 kg **b** 990 kg **c** 1200 kg **d** 3 tonne

Feral camels wander widely according to the conditions. Sometimes they cover 70 kilometres in a single day. They would only travel half this distance if they were working in a camel train, carrying large loads.

- 12** How long would it take a feral camel to travel:
a 140 km? **b** 350 km? **c** 700 km? **d** 820 km? **e** 1500 km?
- 13** How far could a working camel walk in:
a 4 days? **b** 5 days? **c** 12 days? **d** 3 weeks? **e** 5 weeks?
- 14** Large camel trains consist of about 65 camels and 5 camel handlers. A large bull camel is expected to carry up to 600 kg and small camels to carry about 350 kg. How much weight could these camels carry, in tonnes?
a 4 bulls **b** 7 bulls **c** 6 small camels
d 4 bulls and 3 small camels **e** 3 bulls and 5 small camels
- 15** All camels moult in spring and grow a new coat by autumn. Camel hair is used for high-quality coats, garments and artist's brushes. A camel can shed 2.25 kilograms of hair at each moult.
a How much hair is lost by 60 camels at a single moult?
b How many camels are needed to produce 40 kg of camel hair after one moulting?



A snake is a long reptile, covered in scales. All snakes are carnivores. Snakes do not have eyelids, limbs, external ears or legs. While venomous snakes comprise a minority of the species, some possess potent venom that is capable of causing painful injury or death to humans. However, venom in snakes is primarily used for killing or subduing prey, rather than for self defence.

Scorpions are eight-legged carnivorous arthropods and there are about 2000 species of scorpions. Most scorpions are relatively harmless to humans, with their stings only resulting in pain, numbness or swelling. However, a few scorpion species, mostly in the family Buthidae, can be dangerous to humans.

- 1 Find the meaning of each word listed below.
- a venom b toxic c neurotoxin

Snake venoms are like soups. Most of them contain many different components and some of these are toxic to people. For example, in the venom of the Common Tiger Snake, there are at least 6 neurotoxins which affect the human nervous system.

- 2 The table lists the relative toxicity of snake venoms. Plot a picture graph, using snakes as columns, to compare the toxicity of the venoms.

- 3 How many times more toxic is the venom of:

- a a Common Brown Snake than the Common Tiger Snake?
- b an Inland Taipan than the Red-bellied Black Snake?
- c a Taipan than the King Cobra?
- d a Common Tiger Snake than the Western Brown snake?

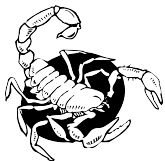
Source of venom	Relative toxicity
Inland Taipan	50.0
Common Brown Snake	12.5
Taipan	7.8
Common Tiger Snake	4.2
Western Brown Snake	1.5
Copperhead	1.0
Red-bellied Black Snake	0.2
Indian Cobra	1.0
King Cobra	0.3
Eastern Diamond-back Rattlesnake	<0.1

- 4 The moulting of a snake’s skin serves two main functions – to replace the old and worn skin and to help get rid of parasites such as mites and ticks. Moulting is repeated periodically throughout a snake’s life. An older snake might shed its skin twice a year. A younger, still-growing snake might shed four times a year. If a snake reaches maturity in 2 years and lives for 10 years, how many times is it likely to shed its skin?

- 5 Pythons are a family of non-venomous snakes. Among its members are some of the largest snakes in the world. Adults range in size from 0.5 m to 10 m in length. The Reticulated Python holds the record of 10 m for the world’s longest snake. Change these Python lengths to the units shown.

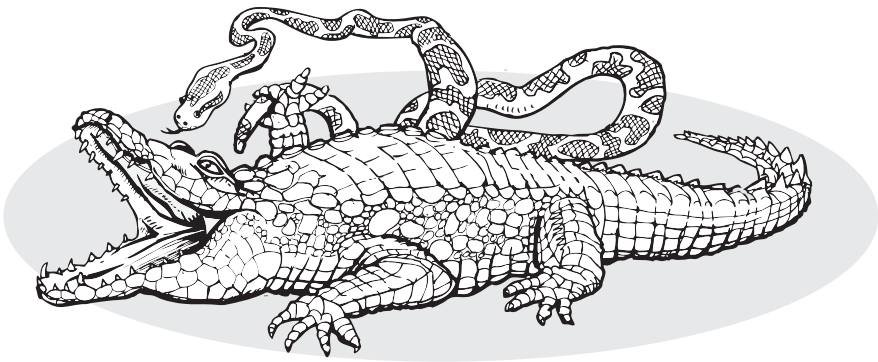
- a 5 m = ____ cm b 8 m = ____ cm c 0.5 m = ____ cm d 0.5 m = ____ mm
- 6 In all types of Carpet Python, there is an average sex ratio of 5 males to every female. On average, how many males would there be in each of these groups of snakes?
- a 24 Pythons b 36 Pythons c 48 Pythons d 42 Pythons e 84 Pythons

- 7 Burmese Pythons are kept by some people as pets. Irresponsible owners released their snakes into the Everglades National Park when they grew too large to manage. In 2005, 190 Pythons were caught by state authorities. In 2006, more than 420 were captured. If the rate of increase in capture rates remained the same, how many Pythons were likely to be caught in 2007?



8 Many Pythons have been observed in epic contests with alligators. The largest Python caught so far in Florida measured five metres and weighed 70 kilograms. Change these Python measurements to the units shown.

- a 500 cm = ____ m
- b 70 kg = ____ g
- c 8 000 g = ____ kg
- d 325 cm = ____ m
- e 33 400 g = ____ kg



9 Despite their awesome size, Pythons are not poisonous. They are excellent swimmers and can cover great distances in small amounts of time. Scientists trapped some Pythons and equipped them with radio transmitters. These Pythons swam 60 kilometres in three hours. Calculate the average swimming speed of Pythons that swim:

- a 30 km in 3 hr b 60 km in 3 hr c 80 km in 3.5 hr
- d 70 km in 3.2 hr e 50 km in 2 hr 24 min

10 Scorpions prefer to live in areas where the temperature ranges from 20°C to 37°C. They can survive the freezing temperatures and scorching heat of the deserts. Write a scale on the arrow below and mark in the scorpion's preferred living temperature.



11 The table at the right lists the sizes of different species of scorpion. Draw a picture graph, using scorpions, to compare the lengths of the species.

Scorpion	Length (cm)
Brown	6
Desert	9.5
Marbled	3
Wood	3.5

12 The Striped Bark Scorpion mates in autumn, spring or early summer. Gestation requires about 8 months and litter sizes vary from 15 to 47.

- a Calculate the average number of scorpions in 4 litters containing 15, 47, 16 and 46 scorpions, respectively.
- b How many scorpions on average would there be in twelve litters? (Use your answer to the previous question as the average.)
- c How many litters of average size are needed to produce 93 scorpions?

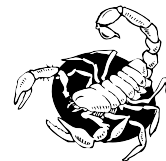
Scorpions take a long time to reach maturity. As they grow, they moult 6 times per year. A scorpion generally lives two to ten years. Some have been recorded as living for up to 25 years.

13 If a scorpion lived for ten years and moulted twice each adult year, how many times would it moult given that it reached maturity in:

- a 4 years? b 6 years? c 5 years 6 months?

14 A scientist recorded the number of times some scorpions moulted until they reached maturity. How long did it take a scorpion to reach maturity if it moulted:

- a 12 times? b 24 times? c 15 times?



The tiger is a mammal of the Felidae family and is native to much of eastern and southern Asia. It can grow up to 4 metres long and weigh up to 300 kilograms. Aside from great bulk and power, the tiger's most recognisable feature is the pattern of dark vertical stripes overlaying near-white to reddish-orange fur.

Whenever you see a colour code letter, find your answer in the shape at the end of the worksheet and colour that section. Colour code: R = red, B = blue, G = green, Y = yellow.

- 1 Find the meaning of each word listed below.

a deciduous	b mangrove	c mutation
-------------	------------	------------

- 2 In the 1920s, about 100 000 tigers existed in the wild. In the 1990s, five different scientists made these estimates of the wild tiger population: 4840, 7040, 6270, 5010 and 6540. Calculate the average estimate of wild tiger numbers.

- 3 Of the tigers surveyed, it was estimated that 3100 of these were Bengal Tigers, 400 were Siberian, 500 were Sumatrans, 1200 were Indo-Chinese, 40 were South Chinas and 700 were Malay Tigers. Calculate the average percentage of these tiger species making up the existing numbers.

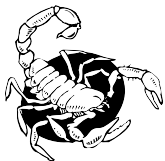
a Bengal	b Siberians	c Sumatrans
d Indo-Chinese	e South Chinas	f Malay

- 4
 - a Calculate the average percentage reduction in tiger numbers from 1920 to 1990.
 - b Suggest why there has been such a decrease in tiger populations.

- 5 The Bengal Tiger is found in parts of India, Bangladesh, Nepal, Bhutan and Burma. It lives in varied habitats: grasslands, subtropical and tropical rainforests, scrub forests, wet and dry deciduous forests and mangroves. Males in the wild usually weigh about 210 kg while the average female weighs about 140 kg. Conservationists have estimated the current number of Bengal Tigers to be 1400 – 40% of their population 10 years ago.
 - a How many Bengal Tigers were there 10 years ago?
 - b How much heavier is the male than the female, as a percentage? (R)

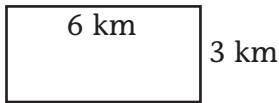
- 6 The Indo-Chinese Tiger is found in Cambodia, China, Laos, Burma, Thailand and Vietnam. These tigers are smaller and darker than Bengal Tigers. Males weigh about 170 kg while females are smaller at 125 kg. Estimates of the total Indo-Chinese Tiger population are 1 500, with about three hundred left in the wild.
 - a What percentage of Indo-Chinese Tigers lives in the wild? (B)
 - b What percentage of Indo-Chinese Tigers lives in captivity? (Y)
 - c How much lighter is the female than the male, as a percentage? (Y)

- 7 Sumatran Tigers are found only on the Indonesian island of Sumatra. They are close to extinction. Sumatran Tigers are the smallest of all living tiger subspecies, with adult males weighing about 110 kg and females about 80 kg. The wild population is estimated to be 450. Habitat destruction is the main threat to the existing tiger population – however, about 60 tigers were recorded as being shot and killed between 1998 and 2000.
 - a What percentage of the wild population of tigers was shot and killed between 1998 and 2000? (G)
 - b How much heavier is the male tiger than the female tiger, as a percentage? (R)

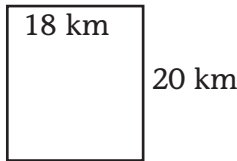


- 8 There is a well-known mutation that produces the white tiger, an animal which is rare in the wild but widely bred in zoos due to its popularity. White tigers can only be born when both parents carry a rare gene. This gene has been calculated to occur in only one in every 10 000 births. What percentage of births is this? (R)
- 9 Adult tigers are fiercely territorial. The size of a tiger's home range mainly depends on prey abundance but also, in the case of male tigers, on access to females. A tigress may have a territory of 20 km² while the territories of males are much larger, covering 80 km² or more. Calculate the area of each tiger territory. Then decide if it belongs to a female or male tiger.

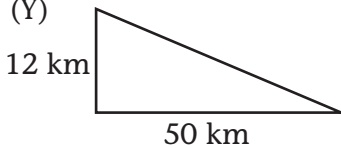
a (G)



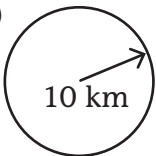
b (B)



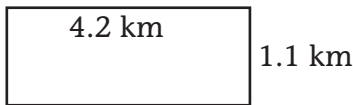
c (Y)



d (G)



e (R)

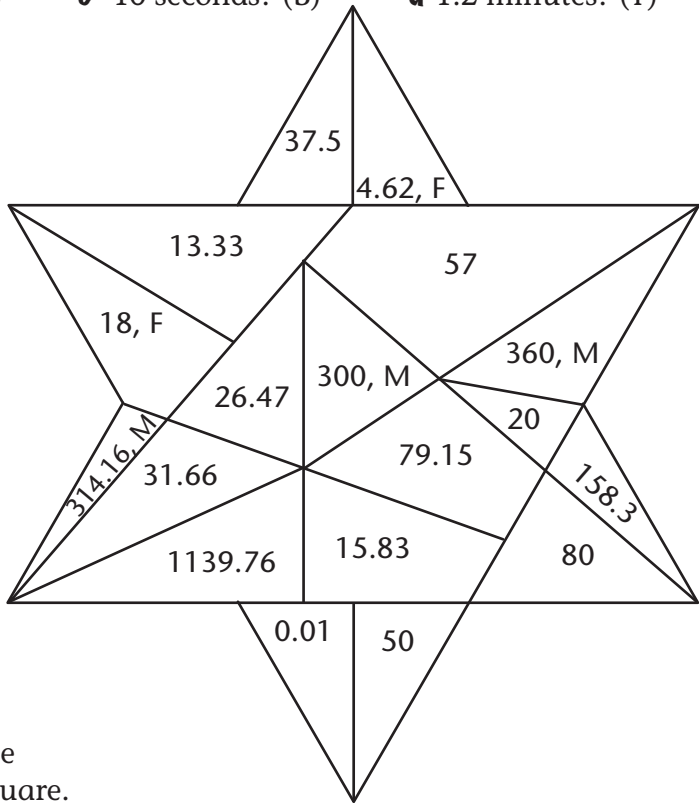


Even with their great masses, tigers can reach speeds of 49 to 65 kilometres per hour. When hunting large prey, tigers prefer to bite the throat and use their muscled forelimbs to hold onto the prey, bringing it to the ground.

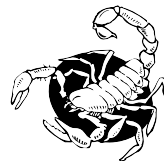
- 10 a Calculate the average speed of these five tigers: 49 km/h, 65 km/h, 53 km/hr, 61 km/h and 57 km/h. (B)
- b Using the average speed you calculated in part a, how many metres can a tiger run in 1 second? (Y)
- 11 On average, how many metres can a tiger run in:
- a 2 seconds? (Y) b 5 seconds? (Y) c 10 seconds? (B) d 1.2 minutes? (Y)



- 12 In the wild, tigers can leap as high as 5 m and as long as 9 m. This makes them one of the highest jumping mammals. The world record for humans is 2.45 m for the high jump and 8.95 m for the long jump. Draw a graph to compare the tiger leaps and human leaps.



- 13 Cut out the 5 shapes you coloured in the diagram. Rearrange them to make a square.



The lion is a member of the family Felidae, and one of four big cats in the genus Panthera. With exceptionally large males exceeding 250 kg in weight, it is the second-largest living cat after the tiger. Wild lions currently exist in Africa and Asia, with a critically endangered population in north-west India. They have disappeared from North Africa, the Middle East and Western Asia.

1 Find the meaning of each word listed below.

- a** defecating **b** synchronise **c** scavenging

Weights for adult lions generally lie between 150 and 250 kg for males, and 120 and 180 kg for females. Head and body length is 170 to 250 cm for males and 140 to 170 cm for females. Shoulder height is about 125 cm in males and 100 cm in females.

2 For these adult lions, calculate the midpoint of the given ranges for the:

- a** weight of a male. **b** weight of a female.
c length of a male. **d** length of a female.

3 Use the midpoints you calculated in the previous question to draw rectangles to represent a male and female lion. Use a scale of 1:100.

Lions spend much of their time resting, and are inactive for about 20 hours per day. Although lions can be active at any time, their activity generally peaks after dusk with a period of socialising, grooming and defecating. Hunting most often takes place at dawn. Each day, lions spend an average of two hours walking and 50 minutes eating.

4 On average:

- a** how many minutes per day are spent resting?
b how many minutes per day are spent walking?
c how many hours per day are spent eating?
d how many seconds per day are spent eating?



5 On average, what percentage of a lion's day is spent:

- a** resting? **b** walking? **c** eating?

Lions can reach speeds of 60 km/h for short bursts. They sneak up to within 30 m of their prey before launching an attack. Lions generally prey on medium size animals such as zebras, gazelle and wildebeest. On average, they are only successful in five out of twenty attempts. A lioness uses her weight and paws to knock her quarry down. She then secures the meal with a bite to the throat or neck, generally suffocating the victim. If a lioness cannot catch her prey in the first few hundred metres, she gives up and chooses another victim.

6 Change these lion speeds to m/s.

- a** 120 km/h **b** 90 km/h **c** 40 km/h **d** 60 km/h **e** 100 km/h

7 At 60 km/h, how long would it take a lion to run these distances to catch its prey?

- a** 20 m **b** 30 m **c** 35 m **d** 40 m **e** 60 m

8 How many times is a lion likely to be successful at catching prey in these numbers of attempts?

- a** 20 **b** 60 **c** 120 **d** 16 **e** 24

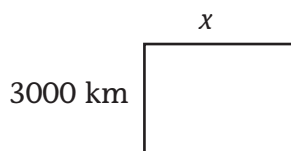


A lion can gorge itself, eating up to 30 kg in one sitting. If it is unable to eat the entire kill, it will rest for a few hours before eating more. On a hot day, the pride may retreat to shade, leaving a male or two to stand guard. An adult lioness requires an average of 5 kg of meat per day, a male 7 kg.

- 9** On average, how much meat will these lions need to eat over the time shown?
a A male, 12 days **b** A male, 3 weeks **c** A female, 23 days **d** A female, 2 months
- 10** On average, how many equivalent 'days of meat' are these amounts for a lioness?
a 30 kg **b** 25 kg **c** 10 kg **d** 12 000 g
- 11** Scavenging makes up about 12% of a lion's food source. Lions listen for vultures and hyenas, which are scavengers, and go to the scene to take what they can for themselves. How much food is likely to be scavenged by these lions during the given time?
a 1 male, 12 wks **b** 2 males, 10 wks **c** 1 female, 16 wks **d** 3 females, 1 yr
- 12** Most lions now live in eastern and southern Africa. Their numbers have rapidly decreased over the last two decades. One current estimate of the African lion population is 31 000, living in the wild. This has decreased from 100 000 in 1990 and 400 000 in 1950.
a Draw a line graph to show the changes in the wild African lion population since 1950.
b Calculate the percentage decrease in the lion population since 1950.
c Suggest why the number of lions has been decreasing.
d From your graph, what was the lion population in 1970?
e From your graph, when was the lion population 150 000?

Like lion numbers, habitat for lions has declined over the past two decades. Since the 1960s, the human population, land cultivation and number of livestock have steadily increased. In 2001, scientists estimated the habitat for lions was about 5 200 000 km², and suggested that this is half of that available in the 1950s. More detailed estimates suggest that the lion habitat is approximately three million km², with about half of this area protected by a national park or reserve. Much of the lion's habitat is fragmented and unprotected.

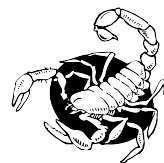
- 13 a** What area of land was available for lion habitat in the 1950s?
b Find the length 'x' of the rectangle below given that its area is 10 400 000 km².



- c** Draw this rectangle to scale on the map of Australia.
d Shade a square of area 3 000 000 km² on the map.
e From detailed estimates, what percentage of the lion's habitat is protected by national parks and reserves?



- 14** Explain the significance of the lion's habitat being fragmented and unprotected.



*The smallest bee is the dwarf bee, **Trigona Minima**, which is about 2.1 mm long. The largest bee in the world is **Megachile Pluto**, which can grow to about 40 mm in length.*

In Australia, the first European Wasps were found in Tasmania in 1959. By 1978 they had also been found in Victoria, South Australia, New South Wales and Western Australia. European Wasps are more aggressive than bees and will attack when their nests are disturbed.

1 Find the meaning of each word listed below.

- a** forage **b** pollen **c** pollination **d** nectar **e** wax

2 Draw pictures of the **Trigona Minima** bee and the **Megachile Pluto** bee, using a scale of 2:1.

Honeybees live in hives or colonies. A small hive contains about 20 000 bees, while larger hives can have 120 000 bees. Hives include one queen, hundreds of drones and thousands of worker bees. The worker bees are female and do not breed. The queen bee is female and creates all the babies for the hive. The drone bees are male and do not have stingers.



3 How many honeybees live in these small hives?

- a** 3 **b** 5 **c** 12 **d** 23 **e** 65

4 How many large hives are needed to support these numbers of bees?

- a** 240 000 **b** 600 000 **c** 200 000 **d** 900 000 **e** 2 million

The worker bees collect pollen and nectar from flowers. The pollen is used as a protein source and the nectar is an energy source. Bees turn the nectar into honey. Workers visit about four thousand flowers to make one tablespoon of honey. Beekeepers are very careful when they remove honey from the hive. They try not to hurt the bees and leave sugar syrup to replace the honey they take.

5 Calculate the quantity of honey, in tablespoons, produced from bees visiting these numbers of flowers.

- a** 8 000 **b** 12 000 **c** 40 000 **d** 15 000 **e** 26 000

6 If a tablespoon of honey weighs 30 g, how many tablespoons make up one kilogram?

7 Calculate the number of flowers visited to make these quantities of honey.

- a** 30 g **b** one kilogram **c** 3 kg **d** 20 kg **e** 170 g

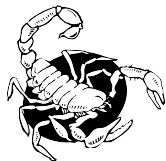
There are about 80 000 commercial bee hives in Queensland. Each hive produces about 70 kilograms of honey. The Queensland bee industry supplies honey, beeswax, honeycomb, queen bee exports and pollination services.

8 Assume that 133 320 flowers produce 1 kg of honey. For a hive to produce 70 kg of honey, how many flowers does a single bee have to visit if, in its hive, there are:

- a** 20 thousand workers? **b** 50 thousand workers?

9 Calculate the annual production of honey in Queensland, in:

- a** kilograms. **b** tonnes.

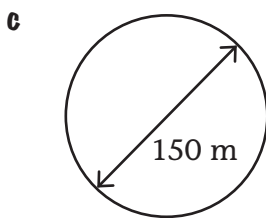
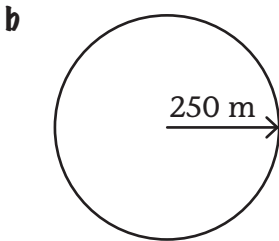
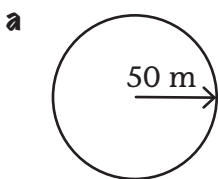


- 10 If honey is valued at \$4 per kilogram, calculate the total value of honey produced.
- 11 One hundred tonnes of beeswax are produced each year. Calculate the value of production at \$4.50 per kilogram.
- 12 Queen bees are valued at \$16 each. If 25 000 are exported, what is the total value of sales?
- 13 Calculate the total value of sales of honey, beeswax and queen bee exports from the bee industry.

In Australia, the temperatures are warm enough for worker European Wasps to survive the winter. This means that if a new queen remains in her old nest, the nest does not begin with a single individual, but with a new queen and several thousand workers. These nests are called ‘overwintering’ nests and by summer can increase in size to house over 100 000 worker wasps.

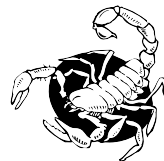


- 14 By summer, how many wasps could be in these numbers of nests?
a 4 b 6 c 8 d 12 e 23
- 15 How many nests are needed to house these numbers of worker wasps?
a 500 000 b 600 000 c 700 000 d 850 000 e 950 000
- 16 Wasp larvae are fed and checked by ‘nursing’ adult workers about 65 times each hour. Total larval development time is about 22 days, depending on the temperature and availability of food.
a Approximately, how many times per minute does an adult worker check the larvae?
b How many times would the larvae be checked over 22 days of development time?
- 17 European Wasps usually forage for food within 50 to 250 metres of their nest, although in some instances they have been recorded flying several kilometres for food. If you have a large number of European Wasps interrupting your outdoor activities, then you probably have a nest nearby. Calculate the total forage area for these spaces.



- 18 The table shows the percentages of deaths from wasp and bee stings.
a Draw a segmented bar graph to compare the death rates of the countries.
b Suggest why the percentage of deaths from wasp stings is relatively low in Australia.

Country	Cause of death	
	Wasps %	Bees %
Sweden	90	10
Denmark	63	37
United States	50	50
Australia	20	80

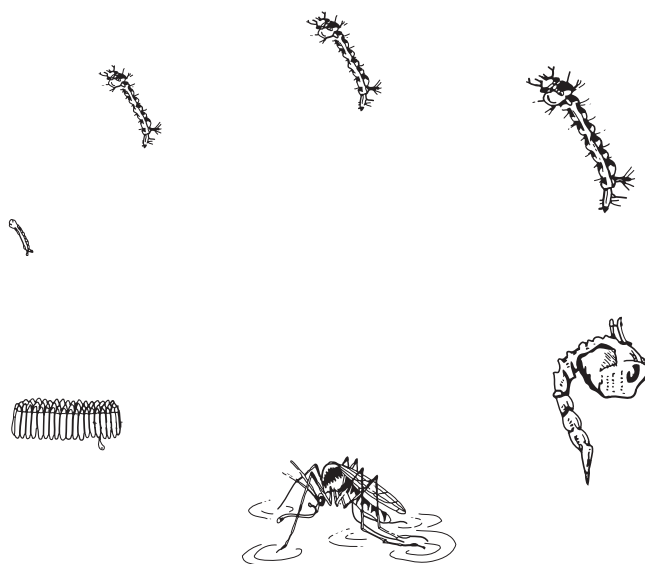


Mosquitoes are insects that make up the family Culicidae. They have a pair of scaled wings, a pair of halteres, a slender body and long legs. The females of most mosquito species suck blood from other animals. This has made them the most deadly disease carriers we know. They have killed millions of people over thousands of years and continue to kill millions per year by spreading diseases.

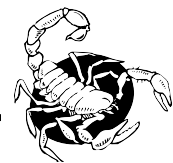
- 1 Find the meaning of each word listed below.
a halteres **b** pupae
- 2 Mosquitoes can grow to 16 mm in length and 2.5 mg in weight.
a How long can mosquitoes grow, in centimetres?
b How many mozzies are needed to make up 1 gram of mozzie flesh?
- 3 A mosquito can fly continuously for 1 to 4 hours at speeds of 2 km/h. They are capable of travelling more than 110 km from their breeding sites.
a How far could a mosquito travel in 4 hours?
b How far could a mosquito travel in one night, without resting? (Allow 6 hours of darkness.)
c How long would it take a mosquito to travel 110 km, without resting?
d How long would it take a mosquito to travel 110 km if it had a 5 km/h assisting breeze and no rests?
- 4 Most species of mosquito feed at night or at dusk or dawn. During the heat of the day, most mosquitoes rest in a cool place but might still bite if disturbed. After drinking blood, adult females lay a raft of up to 400 (depending on the species) tiny, white eggs in still or very slow-moving water. How many eggs could these numbers of mosquitoes lay?
a 5 **b** 6 **c** 18 **d** 23

Most eggs hatch into larvae in about 48 hours. As they grow, they shed or moult their skin about four times, growing larger after each moulting. The larvae eat micro-organisms and organic matter in the water. Mosquito larvae, commonly called 'wigglers', must live in water for about 10 days, depending on the water's temperature. The pupae are lighter than water, and float on the surface while changing into adult mosquitoes over about two days. The newly emerged adult rests on the surface of the water for a short time to allow itself to dry and all its parts to harden before it can fly. A mosquito's adult life is about ten weeks.

- 5 Use the information above to label the diagram of the mosquito life cycle.
- 6 What is the total time, in hours, for a mosquito to complete its life cycle?



Over the past 20 million years or so, mosquitoes have been honing their hunting skills. They are now experts at finding people to bite. Mosquitoes have a battery of sensors designed to track prey. They can sense carbon dioxide and lactic acid from up to 40 metres away.



Certain chemicals in sweat also seem to attract mosquitoes. If you are wearing clothing that contrasts with the background, especially if you move while wearing that clothing, mosquitoes can see you and zero in on you. Mosquitoes detect heat, so they can find warm-blooded mammals and birds easily once they get close enough.



7 Mosquitoes were observed at these distances from a mammal. For each distance, decide whether the mosquito is likely to smell the mammal.

- a 30 m b 45 m c 5 000 cm
d 30 000 mm e 0.05 km f 0.02 km

Repellents like DEET work by disorienting the mosquito as it gets close to its potential next meal, but do not kill mosquitoes. Surprisingly, this works about 95% of the time.

8 How many successes is a mosquito likely to have if it makes these numbers of attacks on a person wearing DEET?

- a 50 b 36 c 120 d 59 e 12 f 250

9 The concentrated mosquito repellent DEET is mixed in with water. Low concentrations of DEET are sufficient for most outdoor protection. A 15% concentration is recommended for most purposes. Which of these repellent mixtures would be satisfactory for most purposes?

- a 50 mL water, 10 mL DEET b 110 mL water, 20 mL DEET
c 120 mL water, 17 mL DEET

In Australia, diseases transmitted by mosquitoes include Dengue fever, Australian encephalitis, Ross River fever and Barmah Forest virus disease. Ross River fever and Barmah Forest virus disease have a wide variety of symptoms, from rashes with fevers to arthritis that can last from months to years. Ross River fever is the most commonly reported mosquito-transmitted disease in all states of Australia. Malaria is rare in Australia.

10 The table at the right shows the number of reported infections of Ross River fever in South Australia.

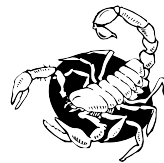
- a Complete the third column of the table, then draw a sector graph to compare the number of infections each year.
b Which year had the most infections?
c Which year had the least infections?
d Suggest an explanation for the nature of the graph you drew.

Year	Infections SA	Sector angle
1992	64	
1993	774	
1994	28	
1995	17	
1996	56	
1997	635	
1998	67	
1999	40	
2000	416	
2001	141	
2002	42	
2003	13	

11 The table below shows the numbers of males and females infected with Ross River fever in South Australia.

- a What is the simplest ratio of infected males to infected females?
b Are males more likely to be infected than females? Explain your answer.

Sex	Number
Male	1182
Female	1110



A bird of prey, or raptor, is a bird that hunts for food primarily on the wing, using its amazing sense of vision. The talons and beaks of raptors tend to be relatively large and powerful, and adapted for tearing or piercing the flesh of their victims.

1 Find the meaning of each word listed below.

a altitude

b juveniles

The Wedge-tailed Eagle is Australia's largest living bird of prey and one of the largest eagles in the world. It reaches about 1.0 m in length and has a wingspan of 2.3 m. Females are larger than males – they average 4.2 kg in weight with males at about 3.2 kg. This eagle has a characteristic long, wedge-shaped tail, and legs that are feathered all the way to the base of the toes.

2 Change these eagle measurements to the units shown.

a 2 m = _____ cm

b 1.0 m = _____ cm

c 2.3 m = _____ cm

d 430 cm = _____ m

e 450 cm = _____ m

Australia's second largest eagle, the White-bellied Sea Eagle, has shorter, more rounded wings than the Wedge-tailed Eagle and no feathers on its lower legs. The White-bellied Sea Eagle is about 0.8 m long with a wingspan of roughly 2.2 m. Males weigh close to 3.1 kg and females 3.5 kg.

3 Change these eagle weights to the units shown.

a 3 kg = _____ g

b 2 kg = _____ g

c 2.3 kg = _____ g

d 4000 g = _____ kg

e 5200 g = _____ kg

Eagles can be seen perched on trees or poles or soaring overhead at altitudes up to 2000 m. An eagle's nest is a large structure of dead sticks, usually reused for years and often reaching considerable sizes. Over many years, a nest can be built up to 1.8 m across, 3 m deep and about 400 kg in weight.

4 Change these measurements to the units shown.

a 2000 m = _____ km

b 2300 m = _____ km

c 600 m = _____ km

d 3 km = _____ m

e 1.6 km = _____ m

f 3000 kg = _____ t

g 3200 kg = _____ t

h 400 kg = _____ t

i 5 t = _____ kg

The diet of the Wedge-tailed Eagle reflects the available prey, but the most important live items are rabbits and hares. Rabbits usually comprise 55% of the diet, but might comprise up to 90%. Other food items include lizards, birds, foxes, feral cats and other mammals. Lambs comprise less than 10% of an eagle's total diet, with most of the sheep meat consumed sourced from carcasses, dying sheep or road fatalities. Less than 2% of all lambs killed are the result of an eagle's attack; foxes and crows account for the remainder.

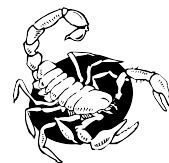
5 The average daily requirement of food for a captive adult eagle is about 600 g.

a If rabbit makes up 55% of the eagle's diet, what is the weight of rabbit needed per day?

b If rabbit makes up 90% of the eagle's diet, what is the weight of rabbit needed per day?

c If 200 lambs are lost to predators, how many are likely to be lost to eagles?

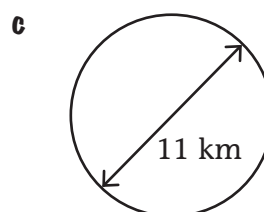
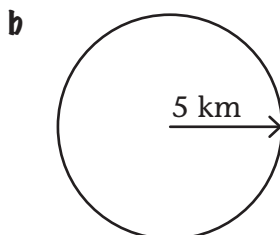
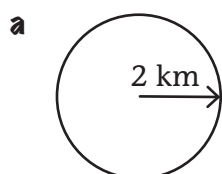
d If 500 lambs are lost to predators, how many are likely to be lost to foxes and crows?



- 6** Under ideal conditions, an eagle can lift about 50% of its body weight. Calculate the weight these eagles can lift.
- a** a female Wedge-tailed Eagle **b** a male Wedge-tailed Eagle
c a female Sea Eagle **d** a male Sea Eagle
- 7** The diet of the Sea Eagle consists mainly of live fish and sea snakes. These two sources account for about 90% of their diet. A Sea Eagle needs approximately 400 g of food per day. Calculate the weight of fish and sea snakes in a Sea Eagle's daily diet.

Each pair of eagles occupies a home range, which may be as little as 9 square kilometres or as large as 120 square kilometres. This large home range is often shared with other eagle pairs for the purpose of hunting.

- 8** Calculate the area of these home ranges.

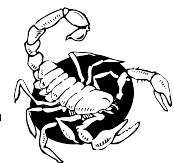


- 9** Assuming its nest is at the centre of the home range, calculate the distance an eagle would have to fly from the nest to reach the edge of these home range areas:
- a** 21 km² **b** 86 km² **c** 120 km²
- 10** The Peregrine Falcon is a very specific hunter that is designed for catching birds in flight, especially starlings, galahs and rosellas. The Peregrine Falcon has a body length of 43 cm and a wingspan of around 100 cm. The female is up to 30 percent larger than the male. Males weigh about 450 g to 750 g, and females weigh 900 g to 1500 g. It is thought that the Peregrine Falcon can dive at speeds of around 360 kilometres per hour.
- a** Calculate the dive speed of the Peregrine Falcon in m/s.
b If a falcon is 200 m above its prey, how long would it take the falcon to dive to its prey?
c If a falcon is 550 m above its prey, how long would it take the falcon to dive to its prey?
- 11** In the wild, the life span of a Peregrine Falcon, is up to 16 years. About 70% of young peregrine falcons die in their first year.
- a** If 200 Peregrine Falcons are born, how many are likely to reach their second year?
b If 120 Peregrine Falcons reach their second year, how many were probably born?

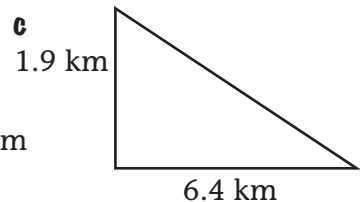
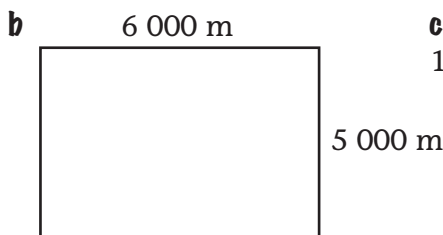
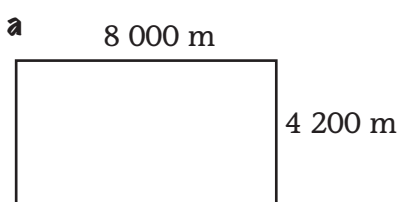


The European Red Fox was introduced to Victoria by fox hunters in 1871 and declared a pest in that state by 1901. Foxes crossed the border into NSW around 1893 and were declared a pest there soon after. They had spread through most of Australia by 1930.

- Commercial harvesting, including hunting, has not been effective in reducing wild rabbit numbers to protect natural resources. Shooting may remove a portion of the population but these are quickly replaced by reproduction. Rabbit shooters harvest only 1.5% of the rabbit population. Even after a 70% reduction in population size, rabbits can regain their former numbers within a single year.



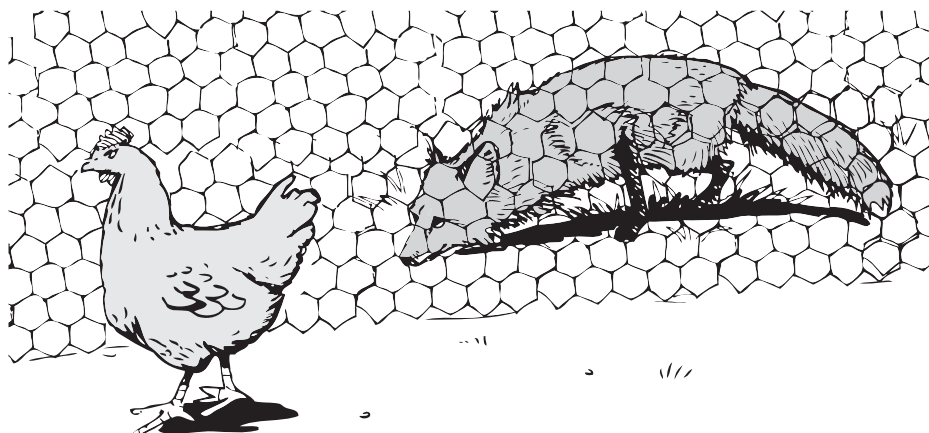
- 7 If the current population of rabbits is 250 million, calculate the number of rabbits:
- a that rabbit shooters harvest per year.
 - b collected to reduce their population by 70%.
- 8 Male foxes weigh 6 kg on average, and female foxes weigh about 5 kg. Red Foxes have small stomachs for their size and can only eat about 10% of their body weight in food each day. This is about half as much as a domestic dog eats. Calculate the amount of food, in grams, that a:
- a vixen (female fox) would eat.
 - b dog (male fox) would eat.
- 9 The typical fox family 'home range' is from 30 to 1 000 ha, with a typical fox density of 6 foxes per 600 hectares. Calculate the number of pairs of foxes that would typically be on these properties. (1 ha = 10 000 m²)

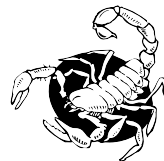


- 10 Researchers found that death rates due to foxes for the Little Tern, Pied Oystercatcher, Beach Stone-curlew and Plover at 20 protected breeding sites were 8%, 11%, 25% and 5%, respectively. Some colonies without fox control had death rates of 100%.

Bird	% mortality	Sector angle
Little Tern		
Pied Oystercatcher		
Beach Stone-curlew		
Plover		

- a Complete the table at the right.
 - b Draw a sector graph to display the relative success of fox protection at breeding sites.
- 11 Serious economic damage results from fox kills of farm livestock, such as lambs, calves, poultry, water fowl and goats. In uncontrolled areas, foxes cause lamb losses of 10% to 30%. Under extreme conditions, lamb deaths can be as high as 50%. One farmer has 90 lambs. Each is valued at \$95. Calculate the value of the losses if foxes kill:
- a 50% of the lambs.
 - b 20% of the lambs.
 - c 10% of the lambs.





*The Cane Toad (*Bufo marinus*) is also known as the Giant Neotropical Toad or Marine Toad. It is a large, terrestrial true toad native to Central and South America. Cane Toads were introduced to Australia to eat cane beetles. The 'whitegrub' larvae of these beetles eat the roots of sugar cane and kill or stunt the plants. About 100 toads were imported from Hawaii to the Meringa Experimental Station near Cairns. The toads bred quickly and more than 3 000 were released in the sugar cane plantations of north Queensland in July 1935.*

1 Find the meaning of each word listed below.

- a parotid b arthritis

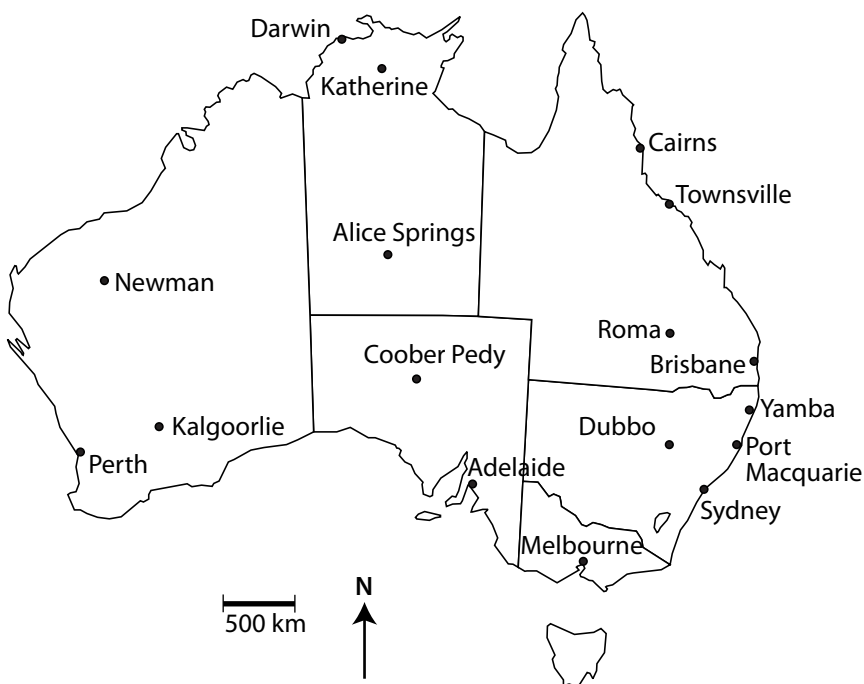
By 2002, Cane Toads had spread throughout the eastern and northern half of Queensland. They have now extended their range to the river catchments surrounding Kakadu National Park, near Katherine, in the Northern Territory. In New South Wales they occur on the coast as far south as Yamba and there is an isolated colony near Port Macquarie.

2 From the map, how far is it

- a Perth to Newman?
b Brisbane to Cairns?
c Melbourne to Sydney?
d Melbourne to Brisbane?
e Adelaide to Darwin?

3 From the map, what is the direction from:

- a Darwin to Katherine?
b Adelaide to Melbourne?
c Perth to Kalgoorlie?
d Brisbane to Roma?
e Dubbo to Melbourne?



4 Shade the map to show the the Cane Toad now inhabits.

5 Use the map to work out how far the toads have spread:

- a in a westerly direction. b in a southerly direction.



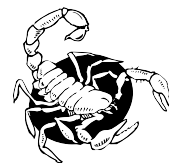
During the day and in cold or dry weather, Cane Toads shelter in moist crevices and hollows, sometimes excavating holes beneath logs, rocks and debris. They can survive the loss of up to 50% of their body water, and survive temperatures ranging from 5 to 40°C.

6 How heavy are these toads, in grams, after they lose 40% of their body weight due to dehydration?

- a 3 kg b 2 kg c 2.6 kg d 1.25 kg e 1.45 kg

7 Toads of these weights were found after losing 35% of their body weight. What was the original weight of each one, in whole grams?

- a 2 kg b 3 kg c 2.1 kg d 3 kg 350 g e 4 kg 230 g



Average-sized adult toads are about 15 cm long. The largest female measured in Queensland was 24 cm long and weighed 1.3 kg. Male Cane Toads are smaller and wartier than females. In other countries, Cane Toads have been recorded at 35 cm long and 2.6 kg in weight.

8 Change these toad measurements to the units shown.

- | | | |
|----------------------------|----------------------------|-----------------------------|
| a 15 cm = _____ mm | b 24 cm = _____ mm | c 35 cm = _____ mm |
| d 340 mm = _____ cm | e 160 mm = _____ cm | f 2 kg = _____ g |
| g 1.3 kg = _____ g | h 2.6 kg = _____ g | i 2 300 g = _____ kg |

Juvenile toads lack the adults' large parotid glands, so they are usually less poisonous. Because they lack this key defence, only about 0.5% of those that hatch from eggs survive to reach sexual maturity and reproduce.

9 How many Cane Toads would reach sexual maturity from these clutches of eggs?

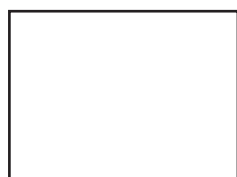
- | | | | | |
|----------------|-----------------|-----------------|-----------------|-----------------|
| a 8 000 | b 20 000 | c 15 000 | d 21 000 | e 35 000 |
|----------------|-----------------|-----------------|-----------------|-----------------|

10 How many eggs are needed to breed these quantities of tadpoles?

- | | | | | |
|--------------|--------------|--------------|--------------|--------------|
| a 200 | b 320 | c 450 | d 600 | e 430 |
|--------------|--------------|--------------|--------------|--------------|

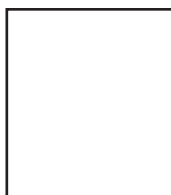
11 Cane Toads occupy more than 500 000 square kilometres of Australia and have reached densities of 2 000 toads per hectare in newly-colonised areas of the Northern Territory. However, the average density of toads in areas where they have been established for more than 20 years is much lower – about 80 toads per hectare in coastal Queensland towns. How many toads would be in the areas below which are near coastal Queensland towns?

a 500 m



400 m

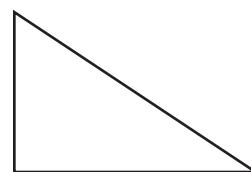
b 400 m



700 m

c

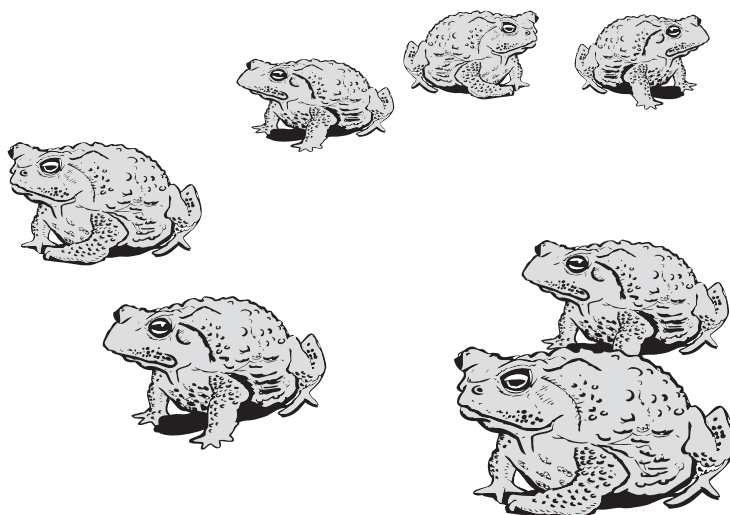
600 m

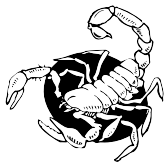


800 m

12 The toads on the western frontier of the Cane Toad advance have evolved larger legs. This is thought to be related to their ability to travel further. As a consequence of their longer legs, larger bodies and faster movement, about 10% of the leading edge of Cane Toads have also developed arthritis. It is estimated that Cane Toads migrate at an average of 40 kilometres per year.

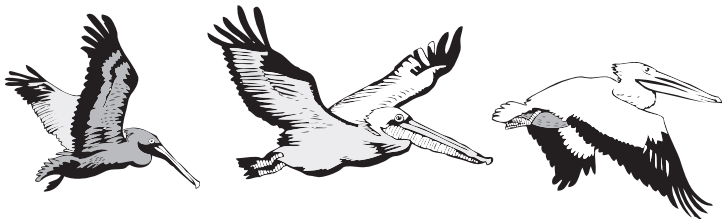
- If there are 25 000 toads leading the movement to the Northern Territory, how many have probably developed arthritis?
- If the toads are now at Katherine, how long will it take them to reach Darwin?
- How long did it take the toads to reach Townsville from Cairns?
- How long would it take the toads to reach Roma from Brisbane?





A pelican is any one of several large water birds with a distinctive pouch under the beak. Pelicans belong to the bird family *Pelecanidae*. Varters, cormorants, gannets, bobbies, frigatebirds and tropicbirds also belong to the pelican family. Pelicans are found on all continents except Antarctica. Swans are large water birds of the family *Anatidae*, which also includes geese and ducks. Swans are closely related to geese, and scientists group the two in the tribe *Cygnini*.

- 1 Find the meaning of each word listed ' '
- a monogamous
 - b gliding
 - c soaring



Pelicans are large birds with enormous, p which can weigh as little as 2.5 kg and have a wingspan of as little as 1.8 m. The largest is believed to be the Dalmatian Pelican, at up to 15 kg with a maximum wingspan of nearly 3.5 m. The Australian Pelican weighs about 5.5 kg and has a wing span of approximately 2.4 m – its 45 cm bill is the longest of any bird.

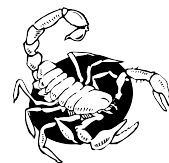


- 2 a Using the scale, label each pelican's wing span with its length.
b Identify and label each pelican.
c The average human has an arm span of 173 cm. Draw a human with outstretched arms beside the pelicans above, using the same scale.
- 3 Convert these pelican measurements to the units shown.
- a 2 kg = ____ g
 - b 2.5 kg = ____ g
 - c 5.5 kg = ____ g
 - d 15 kg = ____ g
 - e 3 000 g = ____ kg
 - f 4 500 g = ____ kg
 - g 3 200 g = ____ kg
 - h 21 000 g = ____ kg
 - i 2 m = ____ cm
 - j 1.8 m = ____ cm
 - k 2.4 m = ____ cm
 - l 45 cm = ____ mm
 - m 500 cm = ____ m
 - n 340 cm = ____ m
 - o 560 cm = ____ m

Pelicans are colonial breeders, with up to 50 000 individuals grouping on islands or secluded shores. Breeding may occur at any time of year, depending on environmental conditions, particularly rainfall. All pelican species lay at least two eggs. Hatching success for undisturbed pairs can be as high as 95 percent, but because of competition between siblings, usually all nestlings but one die within the first few weeks.

- 4 Use the information above to complete the table.

Number of pelicans	Number of pairs	Number of eggs	Number hatched (at 95% success rate)	Siblings surviving
50 000				
40 000				
10 000				
60 000				
30 000				



After about a month, chicks leave their nests to form crèches. After two months, they have learnt to fly and are fairly independent. Wild birds live between 10 and 25 years, or more.

- 5 How many months in:
 - a 2 years?
 - b 25 years?
 - c 12 years?
 - d 10 years?
- 6 Calculate the percentage of their lives spent in the crèche for each of these pelican life spans.
 - a 3 years
 - b 20 years
 - c 10 years
 - d 24 years
- 7 Pelicans have an extremely light skeleton which weighs less than 10% of their total body weight. Calculate the skeleton weight of these pelicans, in grams.
 - a 2.5 kg
 - b 3.6 kg
 - c 15 kg
 - d 5.5 kg
- 8 When fully extended, the bill of a pelican can hold up to 13 litres of water. How many 2 litre milk bottles would this fill?

Pelicans are highly mobile, searching out suitable areas of water and an adequate supply of food. A layer of special fibres deep in the breast muscles holds the wings rigidly horizontal for gliding and soaring. Pelicans are not capable of sustained flapping flight, but can remain in the air for 24 hours at a time. Flight at a height of 1 000 m is common, and heights of 3 000 m have been recorded. By moving from one thermal air current to the next, pelicans can travel long distances with a minimum of effort and reach air speeds of 50 km/h.

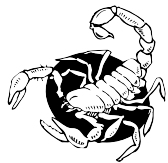
- 9 How far could a pelican travel in 24 hours?
- 10 Change these heights to the units shown.
 - a 1 000 m = ____ km
 - b 4 000 m = ____ km
 - c 4 600 m = ____ km
 - d 2 300 m = ____ km
 - e 3 km = ____ m
 - f 2 km = ____ m
 - g 1.6 km = ____ m
 - h 2.3 km = ____ m
- 11 A mature Black Swan weighs 6 to 9 kg. Its wing span is 1.6 to 2 metres. Calculate:
 - a the average weight of these 6 swans: 6 kg, 9 kg, 8.6 kg, 8.3 kg, 7 600 g, 5 500 g.
 - b the average wing span of these 7 swans: 1.6 m, 2 m, 170 cm, 190 cm, 1.5 m, 175 cm, 215 cm.

A Black Swan nest is essentially a large heap or mound of reeds, grasses and weeds. It is about 1.2 m in diameter and up to 1 metre high, and positioned in shallow water or on an island. The nest is re-used every year, restored or rebuilt as needed.

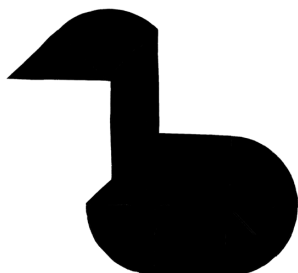
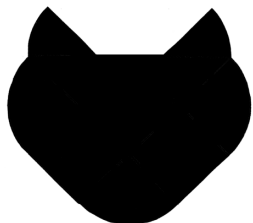
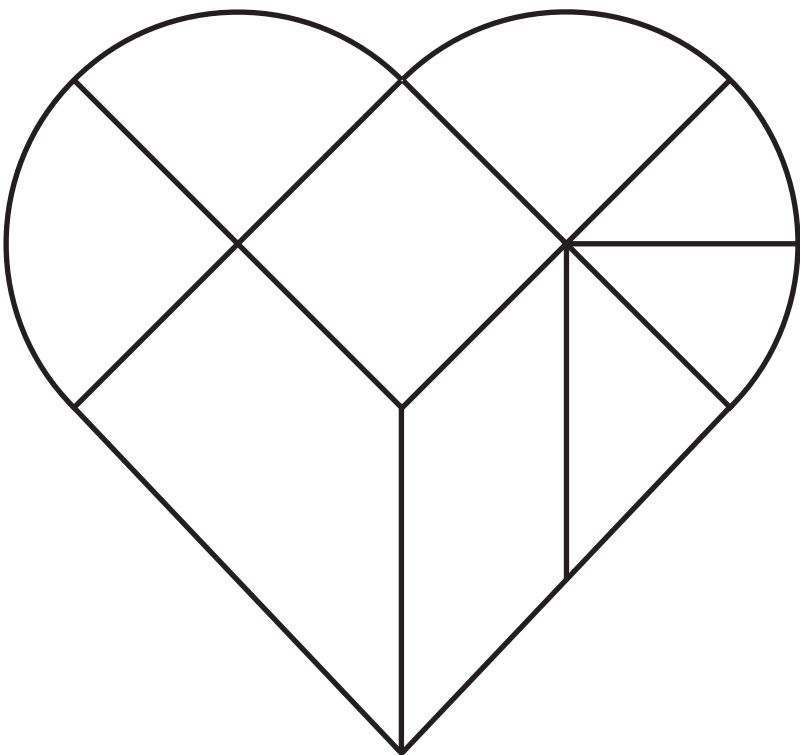
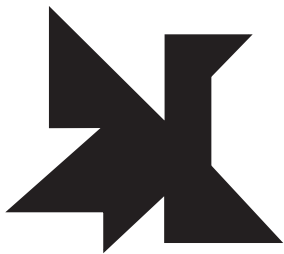
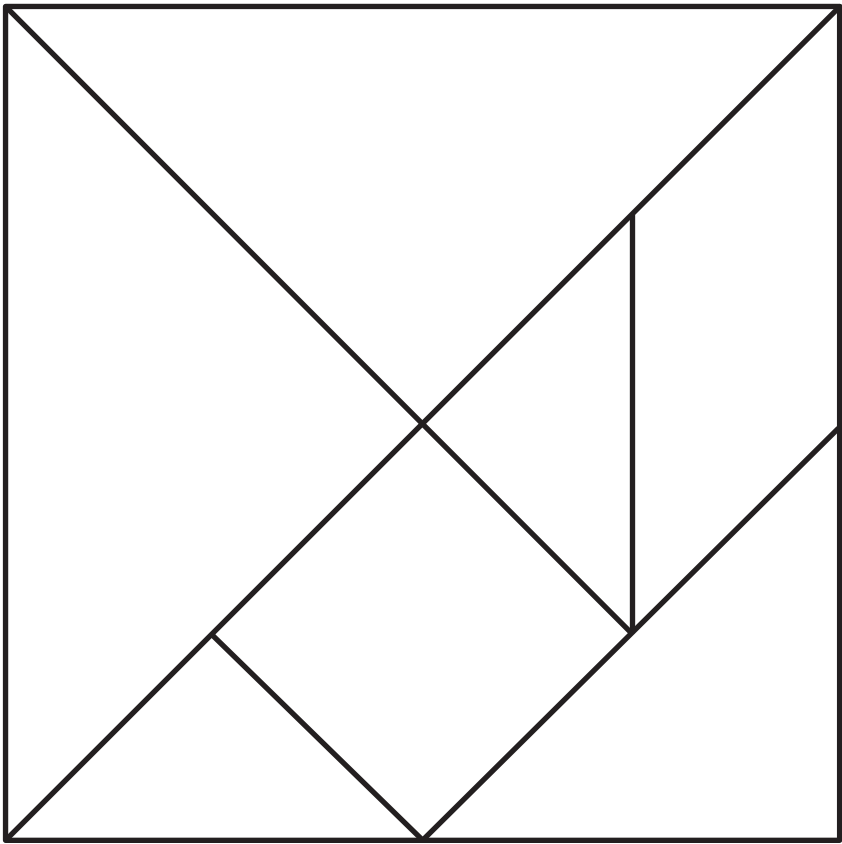
- 12 Calculate the volume of reeds used in a swan's nest.
- 13 Like other swans, the Black Swan is largely monogamous and pairs for life. However, they do have an average divorce rate of 5%. Calculate the average number of pairs of swans divorcing in:

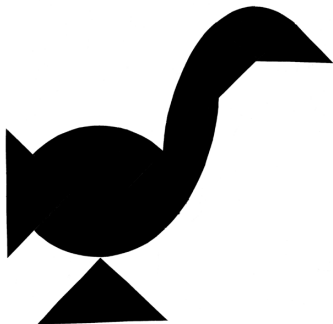
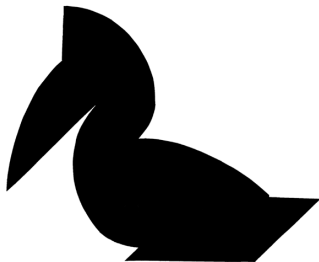
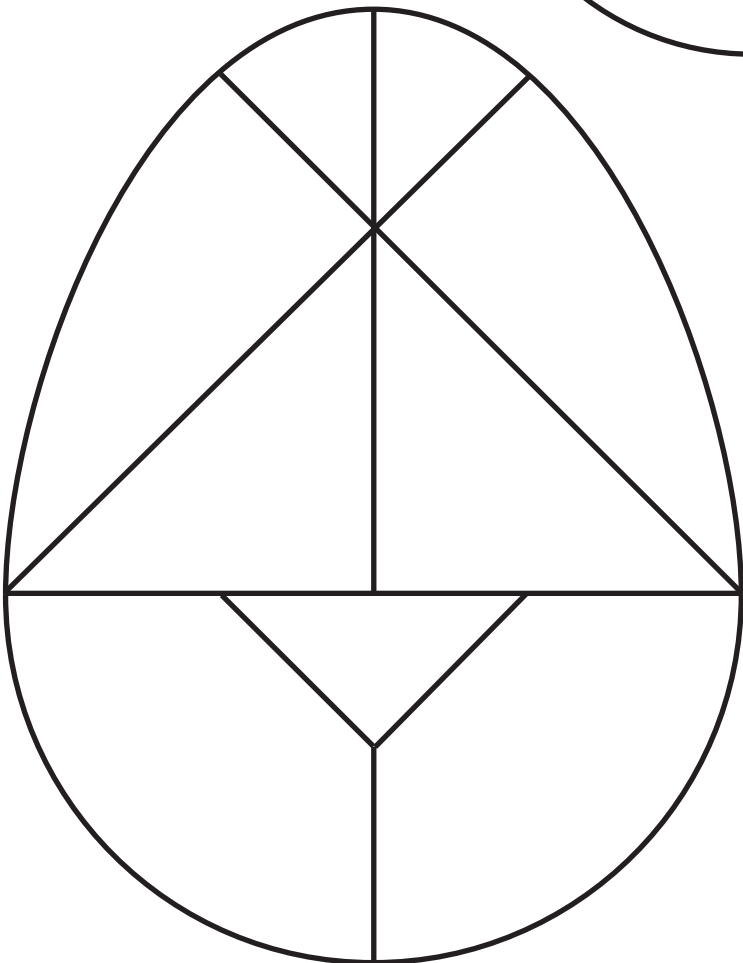
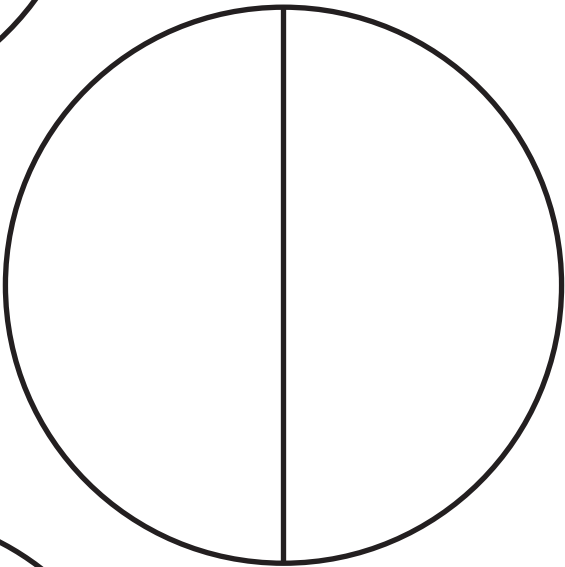
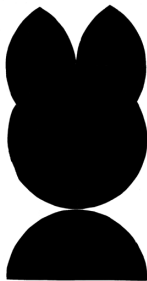
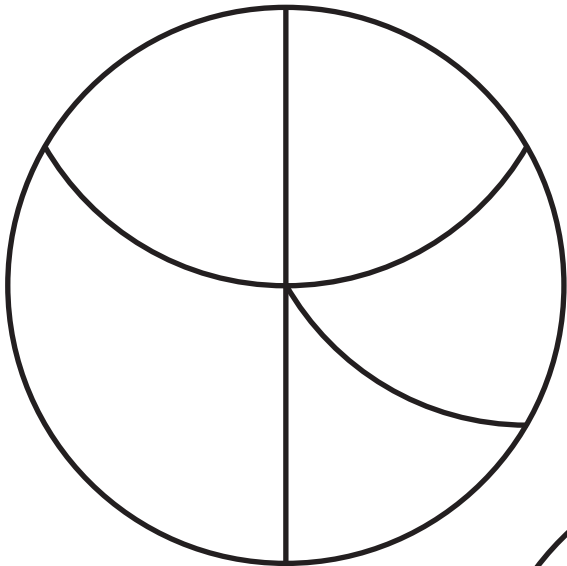
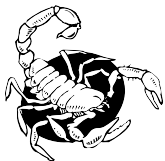
- a 340 pairs.
- b 280 pairs.
- c 120 pairs.
- d 360 pairs.





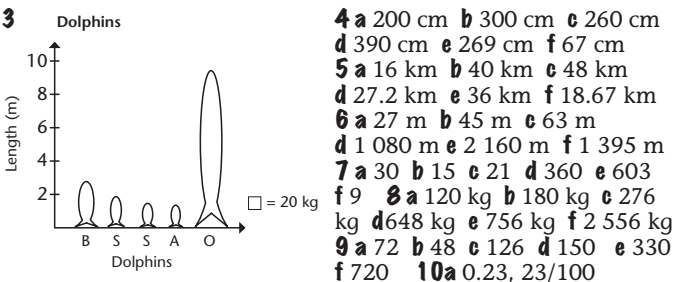
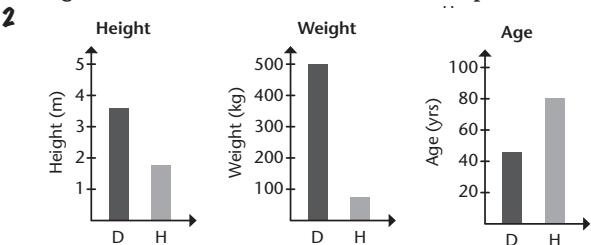
- Cut out the puzzle pieces and use them to make each ‘animal’ shape.
- Write animal names for your shapes.
- Then try making different animal shapes with each set of pieces.





1 - Bottled, spotted and spinners - dolphins

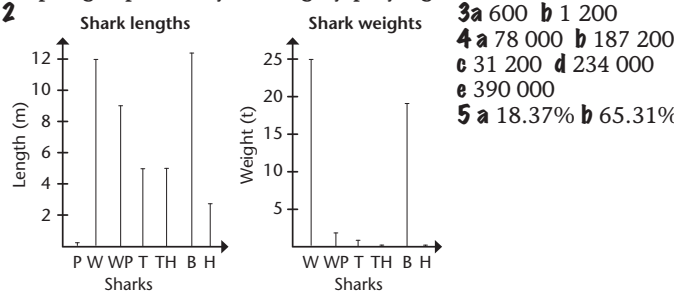
1 a carnivore – an animal that feeds on flesh b breach – the action of launching partially or completely out of the water c sloughed – cast off or shed d conical – shaped like a cone



4 a 200 cm b 300 cm c 260 cm d 390 cm e 269 cm f 67 cm
5 a 16 km b 40 km c 48 km d 27.2 km e 36 km f 18.67 km
6 a 27 m b 45 m c 63 m d 1 080 m e 2 160 m f 1 395 m
7 a 30 b 15 c 21 d 360 e 603 f 9 8 a 120 kg b 180 kg c 276 kg d 648 kg e 756 kg f 2 556 kg
9 a 72 b 48 c 126 d 150 e 330 f 720 10 a 0.23, 23/100
b 0.30, 3/10 c 0.45, 9/20 d 0.04, 1/25 e 0.07, 7/100 11 a 2 b 1 c 3 d 9 e 7 f 10 12 Lack of food, illegal capture, human activity, damage to environment (oil spills, etc.) 13 a 4 b 6 c 9 d 48 e 168 f 4380 14 9 (once every 18 hr)

2 - Fish 'n' chips - sharks

1 a cartilaginous – having a skeleton of cartilage rather than bone b dermal – referring to the skin c denticle – small tooth or tooth-like projection d embryos – unborn or unhatched offspring e predatory – living by preying on other animals



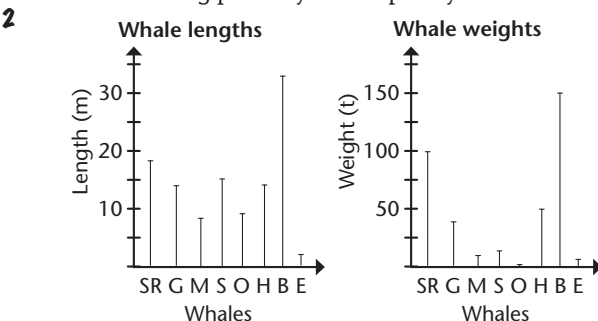
6

Av/yr	%	p/360°
0.9	0.02	0.09
1.3	0.04	0.13
2.1	0.06	0.21
2.5	0.07	0.25
8.9	0.24	0.88
340	9.33	33.57
3290	90.24	324.88

b 13.89 9 0.2 10 a 331 b 768 11 a 146 000 b \$94.73

3 - Giants of the ocean - whales

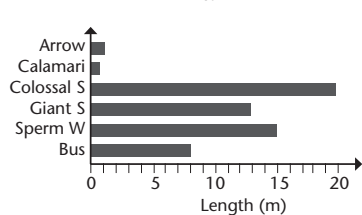
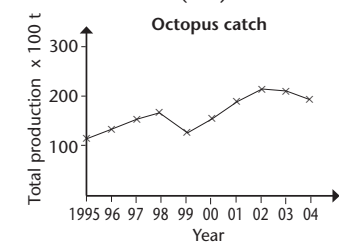
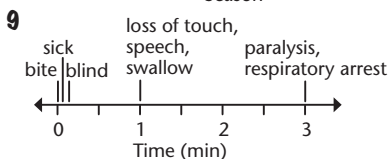
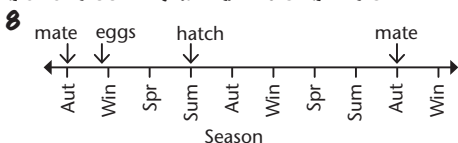
1 a weaning – removing young animals from their source of milk b lunge – a sudden forward movement c breach – the action of launching partially or completely out of the water



3 a 1 040 b 6 000 c 1142 4 Better fishing methods, faster boats, more fishermen. 5 a 9 166.67 kg b 20.37%
6 a 400 b 1600 c 38.10 d 400% 7 a 75 years b 42 c It takes a long time for whales to breed, and a long time to reach sexual maturity. They also breed 1 at a time.
8 a 219 000 L b 219 000 kg c 1 369.86 kg d 1 826 e 547.5 t
9 a 7.2 m, 5.6 m, 3.4 m, 6 m, 5.6 m, 3.8 m, 13.2 m b 36
10 110 t, 38.5 t, 11 t, 27 t, 55 t, 0.48 t, 165 t

4 - Mighty molluscs - octopus, squid & cuttlefish

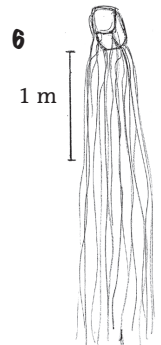
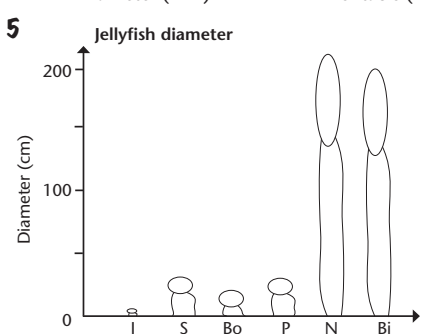
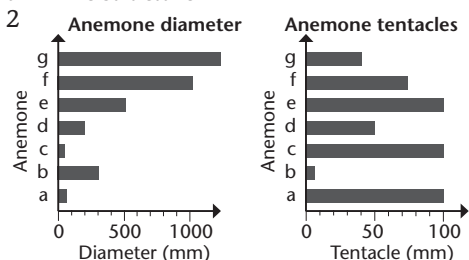
1 a lair – den or hiding place of an animal b larval – juvenile form of an insect undergoing change c respiratory – related to breathing d chameleon – an animal with the ability to change colour 2 a 40 b 56 c 200 d 424 e 392 3 a 2 b 3 c 13 d 27 e 70 4 a 1 000 000 b 1 400 000 c 1 800 000 d 3 000 000 e 4 800 000 5 a 2 b 3 c 1 d 5 e 25 6 a 156 b 208 c 390 d 546 e 884 7 a 1 b 2 c 3 d 4 e 3



12 a 26!!40 b U/25 c U/25 d 26!!40 e U/25
13 a 240 g b 400 g c 480 g d 1200 g e 1920 g 14 a 438 g b 325 g c 1500 g d 1750 g e 1375 g 15 a 1000 b 2400 c 3000 d 20 000 e 102 000

5 - Flowers of the sea - anemones & jellyfish

1 a transparent – easily seen through b nauseous – stomach distress with an urge to vomit c knot – a unit of speed equalling one nautical mile per hour d tentacle – a long, thin, arm-like structure

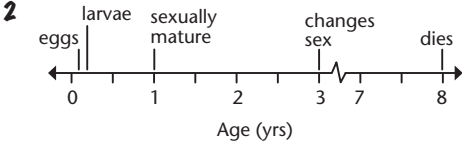


Answers (cont'd)

7 a 24 b 26 c 84 d 30 e 54 8 a 19.17% b 4.17% c 1.67%
d 36.67% e 38.33% 9 a 6 t b 7.5 t c 120 t d 9.75 t e 33 t
10 a 2 b 3 c 8 d 20 e 18

6 - Marinara on the menu - oysters & scallops

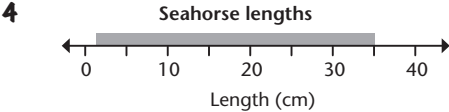
1 a salinity – the amount of salt present in water
b hessian – a type of canvas with a coarse, thick weave



3 a 11 000 b 22 000 c 22 d 33 e 1 4 a 1:10 Yes b 1:50 No
c 1:80 No d 1:30 Yes e 1:45 No 5 a 20 L b 30 L c 65 L d 55 L
e 64.17 L 6 a 5 h b 7 h c 6 h 30 m d 12 h 30 m e 24 h 6 m
7 a 2 b 5 c 10 d 9 e 45 8 a 3 d, Yes b 9 d, Yes c 13 d, No
d 11 d, No e 12 d, No 9 a 3.10 oysters/sec b 24 dozen
10 a 562 500 b 1 218 750 c 2 400 000 11 a 6 750 000,
14 625 000, 28 800 000 b 1 125 000, 2 437 500, 4 800 000
c 83.33% 12 Starfish have eaten many of the scallops.

7 - Weedy dragons & wet horses - seahorses

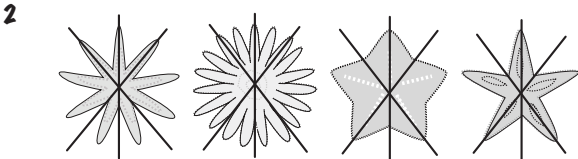
1 a 17 b 2 c 32 2 100 m² 3 a 4 b 3 c 10



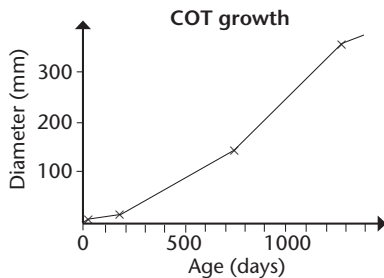
5 a 10 000 b 12 000 c 24 000 d 30 000 e 46 000 6 a 12 b 13
c 14 d 7 e 5 7 0.5% 8 a 8 b 11 c 1 d 15 e 21 9 a 54 794
b 328 767 c 1 666 666 d 1 153 846 e 1 538 461 10 a 767 d
b 9 d c 6 d d 292 d e 456 d 11 a 210 b 350 c 11.67 d 12 600
e 5 040 12 a 98 bpm b 70 bpm c 75 bpm d 74.07 bpm e 150 bpm
13 a 15 000 b 63 000 c 84 000 d 182 500 e 250 14 a 2.33 d
b 3.33 d c 2.67 d d 8.33 d e 11.33 d

8 - Exciting echinoderms - starfish

1 a radiate – to spread out from the centre b regenerate – to
regrow, renew or restore a portion of the body c spawning – a
term used for breeding d buoyant – tending to float on a liquid or
rise in a gas.



3 a 360 b 170 c 1170 d 6 099 e 225 f 10 773 4 a 27 b 63
c 410 d 520 e 780 f 560 5 a 7 200 cm/h b 72 m/h c 0.072
km/h 6 3.57 m/h 7 a 10.68% b 9.89% c 8.93% d 10.06%
e 7.90% f 9.79%
8 a 49 536.63 cm²
b 5 m² c 2.24 m
9 a 200 b 2 000
c 5 000 000
10 at right
11 a ≈ 240 mm
b ≈ 90 mm c ≈ 890 d
d ≈ 1150 d

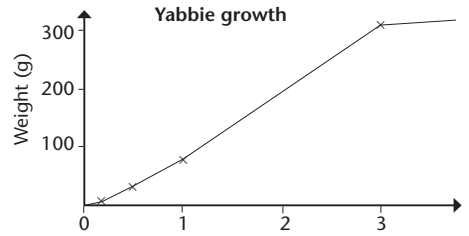


9 - Native fish - Golden Perch & Murray Cod

1 a 3 000 g b 9 000 g c 9 800 g d 8450 g e 5 kg f 7 kg g 6.5 kg
h 9 kg 450 g i 70 cm j 76 cm k 45 cm l 560 cm m 500 mm
n 5 000 mm o 6 500 mm p 456 cm 2 a 1 300 wk b 572 wk
c 127 wk d 220 wk e 21 d f 357 d g 166 d h 90 d 3 a 500 000 m
b 600 000 m c 450 000 m d 1 000 000 m 4 a 1 500 m b 1 750 m
5 a 80 000 b 90 000 c 120 000 d 200 000 e 270 000
6 a 630 000 b 540 000 c 1 170 000 d 1 890 000 e 3 690 000
7 a 120 h b 312 h c 219 h d 206 h e 1 d f 3 d g 1.5 d h 3 d 8 h
8 a 4 000 m b 50 000 m c 120 000 m d 45 700 m e 4 km
f 30 km g 3.4 km h 12.3 km

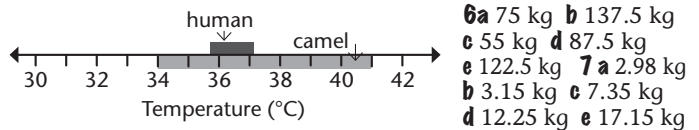
10 - A cute crustacean - yabbies

1 a 28 g b 49 g c 35 g d 81.2 g e 25.2 g 2 a 5 500
b 11 000 c 16 500 d 13 200 e 22 000 3 a 1 000 b 2 000
c 3 000 d 2 400 e 4 000 4 see below
5 a ≈ 260 g
b ≈ 2 yr 6 a 1:250
b 1:1000 c 1:100
d 1:900
7 a, b and d are
okay, as < 1:125
8 a 6 g b 12 g
c 22.5 g d 41.25 g
e 45 g 9 a \$4 025
b \$5 230 c \$11 745
10 \$51.07 11 a 1 950 kg b 3 250 kg c 7 800 kg d 2 080 kg
e 1 365 kg 12 a 7 t b 10.5 t c 19.25 t d 26.25 t e 9.8 t
13 a 3.5 t b 14 t c 7.92 t



11 - The Camelidae family - camels

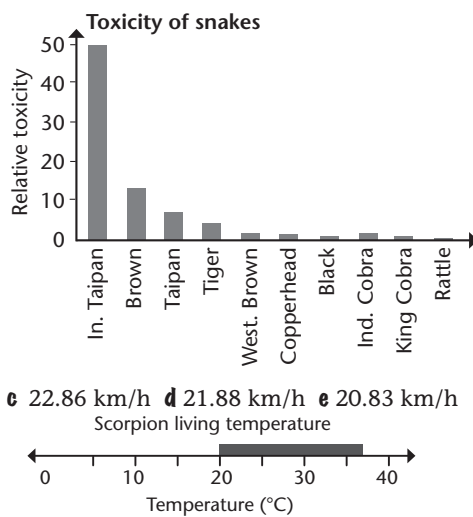
1 a feral – a domesticated animal that now lives in the wild
b dehydrated – condition that occurs when a body loses too
much water c hydrated – caused to absorb water or combine
with water d ingest – to take in food 2 a 2 000 mm
b 3 000 mm c 1 850 mm d 2 150 mm e 400 cm f 300 cm
g 185 cm h 215 cm 3 a 777 000 b 862 470 c 1 062 649
4 a 350 000 b 168 000 c 182 000



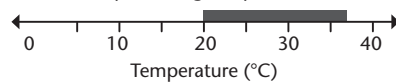
6 a 75 kg b 137.5 kg
c 55 kg d 87.5 kg
e 122.5 kg 7 a 2.98 kg
b 3.15 kg c 7.35 kg
d 12.25 kg e 17.15 kg
8 a 400 L b 600 L c 66.67 L d 333.33 L e 166.67 L 9 a 6 m
b 12 m c 1.5 m d 1.8 m e 3.75 m 10 a 2 t b 4.55 t c 7.8 t
d 3.6 t 11 a 2 b 3 c 4 d 10 12 a 2 d b 5 d c 10 d d 11.71 d
e 21.43 d 13 a 140 km b 175 km c 420 km d 735 km e 1 225 km
14 a 2.4 t b 4.2 t c 2.1 t d 3.45 t e 3.55 t 15 a 135 kg b 18

12 - Silent & deadly - snakes & scorpions

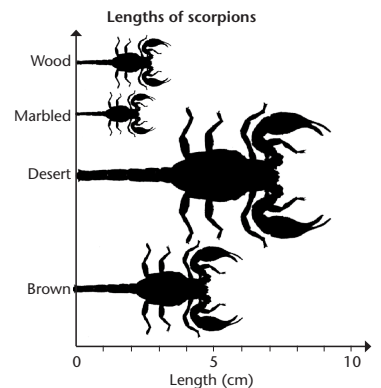
1 a venom – toxin secreted by animals b toxic – harmful to the
body c neurotoxin – a chemical that paralyses nerves



c 22.86 km/h d 21.88 km/h e 20.83 km/h



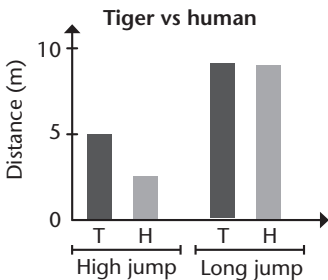
11 see right
12 a 31 b 372 c 3
13 a 36 b 44 c 42
14 a 2 b 4 c 2.5



Answers (cont'd)

13 - Endangered big cat - tigers

1 a deciduous – any plant that sheds its leaves annually
b mangrove – a tropical tree that grows in shallow coastal water
c mutation – any alteration in a gene from its natural state
2 5940 3 a 52.19% b 6.73% c 8.42% d 20.20%
e 0.67% f 11.78%
4 a 94.06% b Loss of habitat, hunting 5 a 3500 b 50%
6 a 20% b 80% c 26.47%
7 a 13.33% b 37.5%
8 0.01% 9 a 18 km² F
b 360 km² M c 300 km² M d 314.16 km² M e 4.62 km² F
10 a 57 km/h b 15.83 m/s
11 a 31.66 m b 79.15 m c 158.3 m d 1 139.76 m
12 at right



14 - Queens & kings of the jungle - lions

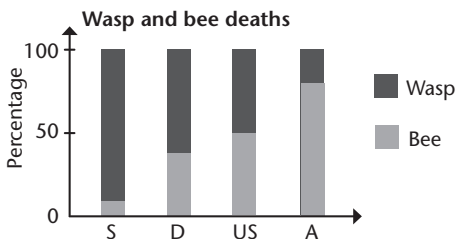
1 a defecating – eliminating solid waste from the digestive tract
b synchronise – to cause two or more events to happen at exactly the same time
c scavenging – searching for and collecting items from discarded waste
2 a 200 kg b 150 kg c 210 cm d 155 cm 3
4 a 1 200 m b 120 m c 0.83 h d 3 000 s
5 a 83.33% b 8.33% c 3.47% 6 a 33.33 m/s b 25 m/s c 11.11 m/s d 16.67 m/s
e 27.78 m/s 7 a 1.20 s b 1.80 s c 2.10 s d 2.40 s e 3.60 s
8 a 5 b 15 c 30 d 4 e 6 9 a 84 kg b 147 kg c 115 kg d 304.17 kg
10 a 6 d b 5 d c 2 d d 2.4 d 11 a 70.56 kg b 117.6 kg c 67.2 kg d 657 kg
12 a
Lion populations

Year	Lion numbers ('000s)
1950	400
1960	300
1970	200
1980	100
1990	50
2000	25

b 92.25% c Loss of habitat, hunting
d ≈250 000 e ≈1983
13 a 10 400 000 km² b 3 466.67 km c
M F Scale 1:100
1 m
e 50% 14 Difficult to mix with other lions, leading to inbreeding.

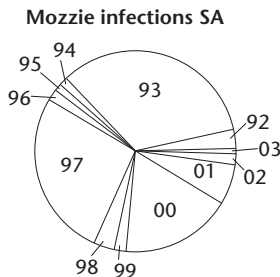
15 - Stings & wings - bees & wasps

1 a forage – search for food and provisions b pollen – the male element in flowering plants
c pollination – transfer of pollen from the male part of the plant to the female part
d nectar – sugar-rich liquid produced by plants to encourage pollination.
e wax – a substance secreted by bees and used to make their honeycombs
2 Trigona Minima should be 4.2 mm long in the drawing, and Megachile Pluto should be 80 mm.
3 a 60 000 b 100 000 c 240 000 d 460 000 e 1 300 000 4 a 2 b 5 c 2 d 8 e 17
5 a 2 b 3 c 10 d 3.75 e 6.5 6 33.33 tablespoons
7 a 4 000 b 133 334 c 400 000 d 2 666 667 e 22 667 8 a 467 b 187
9 a 5 600 000 kg b 5 600 t 10 \$22 400 000 11 \$450 000 12 \$400 000
13 \$23 250 000 14 a 400 000 b 600 000 c 800 000 d 1 200 000 e 2 300 000 15 a 5 b 6 c 7 d 9 e 10
16 a 1.08 b 34 320 17 a 7 853.98 m² b 196 349.54 m²
c 17 671.46 m²
18 a at right b European wasps have not been in Australia for any lengthy period of time.



16 - Blood suckers - mosquitoes

1 a halteres – tiny structures behind the wings of some animals which stabilise them in flight
b pupae – a transformation stage of insects between larva and adult
2 a 1.6 cm b 400 3 a 8 km b 12 km c 55 h d 15.71 h
4 a 2 000 b 2 400 c 7 200 d 9 200 5 egg – 48 h; larvae – 10 d; pupae – 2 d; adult – 10 wks as adult
6 2016 h 7 a 30 m, Yes b 45 m, No c 50 m, No d 30 m, Yes e 50 m, No f 20 m, Yes
8 a 2.5 b 1.8 c 6 d 2.95 e 0.6 f 12.5 9 a 16.67%, Yes b 15.38%, Yes c 12.41%, No
10 a at right b 1993 c 2003
d More rain in some years than others, providing more breeding grounds and therefore more mosquitos.
11 a 197:185 b No, this is nearly 50:50



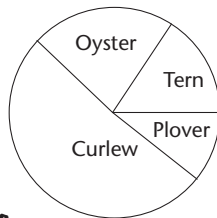
17 - Eagle eyes - birds of prey

1 a altitude – height above the Earth's surface. b juveniles – young people, birds or animals.
2 a 200 cm b 100 cm c 230 cm d 4.3 m e 4.5 m 3 a 3 000 g b 2 000 g c 2 300 g d 4 kg e 5.2 kg
4 a 2 km b 2.3 km c 0.6 km d 3 000 m e 1 600 m f 3 t g 3.2 t h 0.4 t i 5 000 kg 5 a 330 g b 540 g c 4 d 490
6 a 2.1 kg b 1.6 kg c 1.75 kg d 1.55 kg 7 360 g 8 a 12.57 km² b 78.54 km² c 95.03 km²
9 a 2.59 km b 5.23 km c 6.18 km 10 a 100 m/s b 2 s c 5.5 s 11 a 60 b 400

18 - Pesky visitors - rabbits & foxes

1 a colonising – taking over a place for one's own use
b myxomatosis – a viral disease of rabbits c warren – connected underground tunnels occupied by rabbits
2 a 62.4 kg b 72 kg 3 a 42 b 510 4 a 60 000 b 6 000 5 a 320 ha, 160 sheep
b 120 ha, 60 sheep c 306 ha, 153 sheep 6 a \$6 080 b \$2 280 c \$5 814
7 a 3 750 000 b 175 million 8 a 500 g b 600 g 9 a 16 pairs b 15 pairs c 3 pairs
10 a b
Bird mortality rate

Little Tern	8%	59°
Pied Oystercatcher	11%	81°
Beach Stone-curlew	25%	184°
Plover	5%	37°



11 a \$4275 b \$1710 c \$855

19 - Big Bufo Marinus - Cane Toads

1 a parotid – in Cane Toads, this is the gland that secretes a milky poison, known as bufotoxin, when they are in danger
b arthritis – inflammation of joints, causing pain, swelling and stiffness
2 a ≈ 1 100 km b ≈ 1 500 km c ≈ 750 km d ≈ 1 500 km e ≈ 2 600 km
3 a SE b SE c E d W e SW 5 a ≈ 1 400 km b ≈ 2 000 km
6 a 1 800 g b 1 200 g c 1 560 g d 750 g e 870 g
7 a 3 077 g b 4 615 g c 3 231 g d 5 154 g e 6 508 g
8 a 150 mm b 240 mm c 350 mm d 34 cm e 16 cm f 2 000 g g 1 300 g h 2 600 g i 2.3 kg
9 a 40 b 100 c 75 d 105 e 175 10 a 40 000 b 64 000 c 90 000 d 120 000
e 86 000 11 a 1 600 b 2 240 c 1 920 12 a 2 500 b ≈ 10 years c ≈ 13 years d ≈ 15 years

20 - Graceful gliders - pelicans & swans

1 a monogamous – having one mate/breeding partner at a time
b gliding – moving with a smooth, quiet, continuous motion
c soaring – flying or rising high into the air
2 a/b [from left to right] Brown 1.8 m, Australian 2.4 m, Dalmatian 3.5 m
3 a 2 000 g b 2 500 g c 5 500 g d 15 000 g e 3 kg f 4.5 kg g 3.2 kg h 21 kg i 200 cm j 180 cm k 240 cm l 450 mm m 5 m n 3.4 m o 5.6 m
5 a 24 mth b 300 mth c 144 mth d 120 mth 6 a 5.56% b 0.83% c 1.67% d 0.69%
7 a 250 g b 360 g c 1 500 g d 550 g 8 6.5 9 1 200 km 10 a 1 km b 4 km c 4.6 km d 2.3 km e 3 000 m f 2 000 m g 1 600 m h 2 300 m
11 a 7.5 kg b 1.8 m 12 1.13 m³ 13 a 17 b 14 c 6 d 18

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the highest quality learning outcomes with instant usability.

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Maths In the Wild

Did you know that the embryos of the Great White Shark cannabalise each other in the womb because womb space is at a premium? Or that Black Swans are generally monogamous, but do have a 5% divorce rate? What about the fact that a lion spends about 20 hours a day snoozing? The activities in this book provide a wilderness context for practising fundamental mathematics skills. Even your most reluctant student will be motivated and captivated by these real-life topics from the wild. And with all the answers provided, marking will be quick and easy.

Topics include

- Interpreting and drawing graphs

Using and converting units

Speed, time, distance and direction

Percentage

Proportion and rates
- Probability and statistics

Perimeter, area and volume

Symmetry

Calculating costs

Identifying, interpreting and analysing data

Associated titles in the Instant Lessons series

- Maths In series
(Fashion, Science, Technology, Building Design, Crime, Disasters, Food, Sport, the Community, the Environment, the Workplace, Travel)
 - Literacy in Maths (Books 1, 2, 3, 4; Number and Algebra; Data, Measurement and Geometry; A–Z of Maths Terms and Ideas; Maths Across the Curriculum)
 - Instant Lessons in Geometry (Books 1, 2)
 - Assignments in Junior Measurement (Books 1, 2, 3, 4)
 - Assignments in Junior Algebra (Books 1, 2)
- Developing Numeracy series
 - Numbers and the Number System (Books 1, 2, 3)
 - Chance and Data (Books 1, 2, 3)
 - Calculations (Books 1, 2, 3)
 - Algebra (Books 1, 2, 3)
 - Measurement and Space (Books 1, 2, 3)

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series, other Mathematics titles, the full list of contents for all titles and ordering advice.



Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

